Consistency Results

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Summary

Preface

These notes are merely a comprehensive and filtered summary of Kunen's Book - Set Theory: An Introduction to Independence Proofs, along with some of my own intuitions that may be useful to someone, and some additional comments on things I've found elsewhere. Redo

CHAPTER 1

Introduction

All these proofs assume a basic understanding of Set Theory and Logic, mainly basic Model Theory. We also assume some notions of cardinal arithmetic, but mostly cofinality, which I explained in more details here.

You can skip this first subsection and come back when $\sim$ and $\lhd$ were mentioned.

1.1 Some Remarks and Definitions

We will assume for now on that our metatheory is purely finitistic. For this context, Γ will always represent some extension of basic set theory without foundation, then in some sense we can develop at least finite combinatorics in it, and the axioms of Γ will always be decidable.

Definition 1.1

$\Gamma \lhd \Lambda$ iff $\Lambda \vdash \mathrm{Con}(\Gamma)$, and we say Λ is strictly stronger proof-theoretically than Γ.

Clearly $\lhd$ is not reflexive since, by Gödel's second incompleteness theorem, $\Gamma \lhd \Gamma$ iff Γ is inconsistent. Then, let's define a weaker relation:

Definition 1.2

$\Gamma \leq \Lambda$ iff PRA $\vdash \mathrm{Con}(\Lambda) \to \mathrm{Con}(\Gamma)$, such a proof is called a relative consistency proof. Define $\Gamma \sim \Lambda$ iff $\Gamma \leq \Lambda$ and $\Lambda \leq \Gamma$, we say that Γ and Λ are proof-theoretically equivalent.

Clearly $\sim$ is an equivalence relation, and of course the motivation to this notes is to understand this relation (or, at least, $\leq$), i.e., relative consistency proofs over ZFC.

Lemma 1.1

The following is true about $\lhd$ and $\leq$ relations:

1. $\Gamma \lhd \Lambda \Rightarrow \Gamma \leq \Lambda$;
2. $\Gamma \leq \Lambda \ \wedge \ \Lambda \lhd \Theta \Rightarrow \Gamma \lhd \Theta$;
3. The relation $\lhd$ is transitive;
4. $\Gamma \leq \Lambda \ \wedge \ \Lambda \lhd \Gamma \Rightarrow \neg\,\mathrm{Con}(\Gamma) \ \wedge \ \neg\,\mathrm{Con}(\Lambda)$.

Proof. 1. Assume $\Lambda \vdash \mathrm{Con}(\Gamma)$, now, if $\neg\,\mathrm{Con}(\Gamma)$ in PRA, then Λ can also prove that $\neg\,\mathrm{Con}(\Gamma)$ and, therefore we can argument finitistically in PRA that $\neg\,\mathrm{Con}(\Lambda)$.

2. If PRA $\vdash \mathrm{Con}(\Lambda) \to \mathrm{Con}(\Gamma)$ and $\Theta \vdash \mathrm{Con}(\Lambda)$ then, since Θ can do finitistic reason, then $\Theta \vdash \mathrm{Con}(\Lambda) \to \mathrm{Con}(\Gamma)$ also and, therefore, $\Theta \vdash \mathrm{Con}(\Gamma)$, i.e., $\Gamma \lhd \Theta$.

3. Immediate from 1. and 2.

4. By 2. we have $\Gamma \lhd \Gamma$, then $\neg\,\mathrm{Con}(\Gamma)$, therefore, since $\Gamma \leq \Lambda$, $\neg\,\mathrm{Con}(\Lambda)$. ∎

Finally, in order to pin down some notations, we will denote by BST (Basic Set Theory) the following set of axioms: $\mathcal{E}$, $\mathcal{R}$, $\mathcal{C}$, Pairing, $\mathcal{U}$ and $\mathcal{P} \vee \mathcal{F}$.

The last one should seem a bit odd, but it derives from the fact that many of the basic theorems about functions and relations have dual proofs, one using Power Set and one using Replacement.

1.2 Model Approximations of V

Just to pin down some intuition, we will see that WF is the class of all well founded sets which, in some sense, means we can write all it's elements "extensively" until we reach $\emptyset$, so they are $\in$-trees whose height (which we call rank) is not arbitrary large in ON.

In particular, we can then though about initial segments of WF as the sets $R(\gamma)$ whose height (or rank) of its elements are less than γ, i.e., if we write the set extensively, we have less than γ nested sets until we reach $\emptyset$. Or we can talk about initial segments of the form $H(\kappa)$ whose elements has transitive closures of cardinality less than κ, which, of course, does imply that its height is less than κ, and then $H(\kappa) \subseteq R(\kappa)$. At first there is not primary reason to believe that $H(\kappa) = R(\kappa)$, and we will see that this is in fact pretty rare.

1.2.1 $R(\gamma)$

It's common to define $R(\gamma) = V_\gamma$ but this definition requires power set axiom, which could be avoided if we define $R(\gamma) := \{x \in WF : \mathrm{rank}(x) < \gamma\}$, we will not go this way, tho.

To see the equivalence, just note that $b \in R(\alpha+1)$ iff $b \subseteq R(\alpha)$. In fact, if $\mathrm{rank}(b) < \alpha + 1$, then, given $x \in b$, $\mathrm{rank}(x) < \mathrm{rank}(b) \leq \alpha$, i.e., $x \in R(\alpha)$. And, if $b \subseteq R(\alpha)$, then

$$\mathrm{rank}(b) = \sup\{\mathrm{rank}(x) + 1 : x \in b\} \leq \alpha$$

so $b \in R(\alpha+1)$.

Lemma 1.2

$R(\gamma)$ is transitive, for every $\gamma \in ON$.

Proof. Clearly $R(0) = 0$ is transitive. Assume $R(\alpha)$ is transitive, and let $x \in y \in R(\alpha+1) = \mathcal{P}(R(\alpha))$, then $x \in y \subseteq R(\alpha)$, so $x \in R(\alpha)$ which, begin transitive, implies that $x \subseteq R(\alpha)$, therefore $x \in \mathcal{P}(R(\alpha)) = R(\alpha+1)$. Now, if $R(\beta)$ is transitive, for every $\beta < \lambda$, with λ a limit ordinal, then, given $x \in y \in R(\lambda) = \bigcup_{\beta<\lambda} R(\beta)$, $y \in R(\beta)$ for some $\beta < \lambda$ and, since it's transitive, $x \in R(\beta)$ and, therefore, we also have that $x \in R(\lambda)$. ■

The following Theorem is clear by transfinite induction and the definition of $R(\alpha)$:

Theorem 1.1

$|R(\omega+\alpha)| = \beth_0$ for all α, in particular $|R(\alpha)| = \beth_\alpha$ for all $\alpha \geq \omega^2$

It's also not hard to prove that $R(\alpha) \cap ON = \alpha$ using transfinite induction. The base and limit cases are trivial, for the successor case, let $\beta \in R(\alpha+1) \cap ON = \mathcal{P}(R(\alpha)) \cap ON$, then $\beta \subseteq R(\alpha) \cap ON = \alpha$, i.e., $\beta \leq \alpha$ and, since $\alpha \in R(\alpha+1) \cap ON$, then $R(\alpha+1) \cap ON = \alpha+1$.

We will see how this model approximation gives us an independence proof of the infinity and replacement axioms, using $R(\omega)$ and $R(\omega+\omega)$, respectively.

Let's complete this discussion talking about the class WF of all well founded sets.

Since $R(ON) = WF$, we also have that $ON \subseteq WF$, then it's a proper class. Furtherfore, given $x \in y \in WF$, $y \in R(\alpha)$ for some α and, since it's transitive, $x \in R(\alpha) \subseteq WF$, then WF is itself transitive. Note how almost all properties of $R(\alpha)$ and WF are proved by transfinite induction.

1.2.2 $H(\kappa)$

The following will be important to give an independence proof for the power set and union axioms.

Definition 1.3

Given a cardinal κ, let
$$H(\kappa) := \{x \in WF : |\mathrm{trcl}(x)| < \kappa\}$$
In particular, $H(\aleph_1) = HC$ is called the set of hereditarily countable sets.

$H(\kappa)$ is clearly transitive since, if $x \in y$, then $\mathrm{trcl}(x) \subseteq \mathrm{trcl}(y)$.

This set is also way smaller than $R(\alpha)$, as the next theorem shows.

Theorem 1.2

For any infinite cardinal κ, $|H(\kappa)| = 2^{<\kappa}$

Proof. Given $\lambda < \kappa$, if $x \subseteq \lambda$, then $|\mathrm{trcl}(x)| \leq |\mathrm{trcl}(\lambda)| = \lambda < \kappa$, i.e., $\mathcal{P}(\lambda) \subseteq H(\kappa)$, so $|H(\kappa)| \geq 2^{<\kappa}$.

Now, let $F : H(\kappa) \to \bigcup\{\mathcal{P}(\lambda \times \lambda) : \lambda < \kappa\}$ be defined as follows: Given $x \in H(\kappa)$, let $\lambda = |\mathrm{trcl}(x) \cup \{x\}| < \kappa$ and, by AC, choose $F(x) \subseteq \lambda \times \lambda$ such that $(\lambda, F(x)) \cong (\mathrm{trcl}(x) \cup \{x\}, \in)$, which is injective, since x is determined by the order type of $\in$ on $\mathrm{trcl}(x) \cup \{x\}$. ■

1.3 Relations Between R and H

First of all, note the following relation between $H(\kappa)$ and $R(\kappa)$

Lemma 1.3

$H(\kappa) \subseteq R(\kappa)$ for every cardinal κ.

Proof. Let $x \in H(\kappa)$, we will show that $\text{rank}(x) \leq |\text{trcl}(x)| < \kappa$. Consider the rank function restricted to $\text{trcl}(x)$, we claim that its transitive on $\text{rank}(x)$.

Let $\beta \in \text{rank}(x) = \sup\{\text{rank}(y) + 1 : y \in x\}$, then $\beta < \text{rank}(y) + 1$ for some $y \in x$, i.e., $\beta \leq \text{rank}(y)$ for some $y \in \text{trcl}(x)$. If $\beta = \text{rank}(y)$, then $\beta \in \text{Im rank} \restriction_{\text{trcl}(x)}$. Otherwise, $\beta < \text{rank}(y)$, and we can repeat the same process to find that $\beta \leq \text{rank}(z)$, for some $z \in y \subseteq \text{trcl}(x)$. Since this can't continue forever, because $\text{trcl}(x)$ is $\in$-well founded, then eventually we will reach a $\beta = \text{rank}(w)$ for some $w \in \text{trcl}(x)$. ■

As we noted earlier, $H(\kappa)$ is almost always way smaller than $R(\kappa)$, but the following theorem gives us an exact criterion for when they're equal

Theorem 1.3

For cardinals $\kappa > \omega$, $H(\kappa) = R(\kappa)$ iff $\kappa = \beth_\kappa$.

Proof. ($\Rightarrow$) Whenever $\omega^2 \leq \alpha < \kappa$, we have that $R(\alpha) \in R(\kappa) = H(\kappa)$, then, since its transitive

$$|\text{trcl}(R(\alpha))| = |R(\alpha)| = \beth_\alpha < \kappa$$

and, therefore, $\beth_\kappa = \sup_{\alpha<\kappa} \beth_\alpha \leq \kappa$, so, because $\beth_\kappa \geq \kappa$, we have $\beth_\kappa = \kappa$.

($\Leftarrow$) If $x \in R(\kappa)$, then $x \in R(\alpha)$ for some $\omega^2 \leq \alpha < \kappa$, and then $|\text{trcl}(x)| \leq |R(\alpha)| = \beth_\alpha < \beth_\kappa = \kappa$, so $x \in H(\kappa)$, which implies, by the previous lemma, that $R(\kappa) = H(\kappa)$. ■

We can also prove by transfinite induction that $R(\omega) = H(\aleph_0) = HF$, called the set of hereditarily finite sets. Just note that $R(0) \subseteq H(\aleph_0)$ and, if $R(n) \subseteq H(\aleph_0)$, given $x \in R(n+1) = \mathcal{P}(R(n))$, since $R(n)$ only contains finite sets, then $\mathcal{P}(R(n))$ also contains finite sets, therefore x is finite. Furthermore, since $x \subseteq R(n) \subseteq H(\aleph_0)$, then all subsets of x are hereditarily finite, then $x \in H(\aleph_0)$.

To be fair, the jump between $x \subseteq H(\aleph_0)$ and $|x| < \omega$ to $x \in H(\aleph_0)$, although kinda intuitive, implicitly assumed that ω is a regular cardinal. Let's restate the more general case:

Lemma 1.4

Let κ be a regular cardinal. If $y \subseteq H(\kappa)$ and $|y| < \kappa$, then $y \in H(\kappa)$.

Proof. Let's first show that, for any given y, we have that

$$\text{trcl}(y) = y \cup \bigcup_{x \in y} \text{trcl}(x) \tag{1}$$

which follows from the fact that $\bigcup^{n+1} y = \bigcup_{x\in y}\bigcup^{n} x$, for $n \geq 0$. By induction, clearly $z \in \bigcup y$ iff $z \in x$ for some $x \in y$, i.e., $z \in \bigcup_{x\in y} x$. Now, if it's true for n, then

$$\bigcup^{n+2} y = \bigcup\Big(\bigcup^{n+1} y\Big) = \bigcup\left(\bigcup_{x\in y}\bigcup^{n} x\right) = \bigcup_{x\in y}\bigcup^{n+1} x$$

and now we can easily see how (1) follows:

$$\operatorname{trcl}(y) = y \cup \bigcup_{n\geq 0}\bigcup^{n+1} y = y \cup \bigcup_{n\geq 0}\left(\bigcup_{x\in y}\bigcup^{n} x\right) = y \cup \bigcup_{x\in y}\left(\bigcup_{n\geq 0}\bigcup^{n} x\right) = y \cup \bigcup_{x\in y}\operatorname{trcl}(x)$$

Now, for the real proof, since $|y| < \kappa$ and $y \subseteq H(\kappa)$, then $|\operatorname{trcl}(x)| < \kappa$ for every $x \in y$, then

$$|\operatorname{trcl}(y)| \leq |y| + \sum_{x\in y} |\operatorname{trcl}(x)| < \kappa$$

since κ is regular, otherwise κ would be the sum of less than κ sets with size less than κ. ■

CHAPTER 2

Easy Consistency Proofs

This chapter will be devoted to use our model approximations to give "easy" consistency and independence proofs about almost all axioms of ZFC.

Note that hierarchies like $H(\kappa)$ and $R(\gamma)$ are defined in terms of levels of boundedness for some "rank" function ρ. In fact, since almost every ZFC axiom states the existence of some set y we can construct using sets x that already exists, they can be expressed as a $\forall\exists$, and then the proofs will just consists of showing our set y has a bounded rank in terms of the ranks of the sets x.

2.1 First Axioms

Since WF, $H(\kappa)$ and $R(\gamma)$ are very similar, a lot of consistency proofs will also be similar. In order to avoid that we will prove the following useful lemma[1].

Lemma 2.1

For any class M:

1. If M is transitive, then $M \vDash \mathcal{E}$;
2. If $M \subseteq WF$, then $M \vDash \mathcal{R}$;
3. If $\forall z \in M\ \forall y \subseteq z[y \in M]$, then $M \vDash \mathcal{C}$;
4. If $\mathcal{F} \in M[\bigcup \mathcal{F} \in M]$, then $M \vDash \mathcal{U}$;
5. If M is transitive and if for any function f, with $\mathrm{dom}(f) \in M$ and $\mathrm{Im}(f) \subseteq M$, we have that $\mathrm{Im}(f) \in M$, then $M \vDash \mathcal{F}$.

[1] Note that such hypothesis in each item of the following lemma are unnecessary, in the sense that they are way stronger than the conclusion. In fact, they prove a stronger axiom, namely the second order version of it, which says "all" possible subsets satisfying the properties we want exists, and not just the definable ones. We will see how to weaken this later.

Proof. 1. We need to prove that $\forall x, y \in M(x = y \leftrightarrow \forall z \in M(z \in x \leftrightarrow z \in y))$. Clearly, if $x = y$, then $\forall z \in M(z \in x \leftrightarrow z \in y)$. Now, for the other direction, if $z \in x$ or $z \in y$, then, since $x, y \in M$, and M is transitive, $z \in M$, which means $\forall z(z \in x \leftrightarrow z \in y)$, then in fact $x = y$.

2. The Axiom of Regularity (or Foundation) relativized to M is

$$\forall x \in M[\exists w \in M(w \in x) \rightarrow \exists y \in M(y \in x \wedge \nexists z \in M(z \in x \wedge z \in y))]$$

Given $x \in M$, let $\widetilde{x} = x \cap M$. Since $w \in \widetilde{x}$, and $M \subseteq WF$, then there is a $\in$-minimal element y of $\widetilde{x}$ in M. Now, if $z \in M$, and $z \in x$, then $z \in \widetilde{x}$, which means $z \notin y$.

3. Let $\varphi(x, z, \overline{v})$ be a $\{\in\}$-formula without y free. We need to prove that

$$\forall z, \overline{v} \in M \ \exists y \in M \ \forall x \in M[x \in y \leftrightarrow x \in z \wedge \varphi^M(x, z, \overline{v})].$$

Given $z \in M$, let $y = \{x \in z : \varphi^M(x, z, \overline{v})\}$, then $y \subseteq z$ and, by hypothesis, implies that $y \in M$.

With these proofs, it's clear how to do the same for 4. and 5. ■

Let's now see much axioms we can prove in our approximate models

Corollary 2.1

If γ is a limit ordinal, then $R(\gamma) \vDash Z - \mathcal{P} - \mathcal{I}$. Furthermore, $HF = R(\omega) \vDash \mathcal{F}$.

Proof. Since all three approximate models are transitive and subsets of WF, let's omit 1. and 2.

3. Let $z \in R(\gamma)$, then $z \in R(\alpha)$, for some $\alpha < \gamma$. Since λ is a limit ordinal, $\alpha + 1 < \gamma$ and, if $y \subseteq z$, then $y \in \mathcal{P}(z) \subseteq R(\alpha + 1) = \mathcal{P}(R(\alpha)) \subseteq R(\gamma)$, then $y \in R(\gamma)$.

4. Similarly to 3., let $\mathcal{F} \in R(\gamma)$, then $\mathcal{F} \in R(\alpha)$ for some $\alpha < \gamma$. Given $x \in \bigcup \mathcal{F}$, $x \in y$, for some $y \in \mathcal{F}$, but since $R(\alpha)$ is transitive, $x \in R(\alpha)$ and, therefore, $\bigcup \mathcal{F} \subseteq R(\alpha)$, which implies that $\bigcup \mathcal{F} \in \mathcal{P}(R(\alpha)) = R(\alpha + 1) \subseteq R(\gamma)$.

(5. for $R(\omega)$): Since, as we saw, $R(\omega) = H(\omega)$ and ω is regular, then by Corollary 2.1, $HF \vDash \mathcal{F}$. ■

Corollary 2.2

$WF \vDash ZF - \mathcal{P} - \mathcal{I}$

Proof. They all follow from the previous corollary and the fact that $WF = R(ON)$. ■

Corollary 2.3

If κ is an infinite cardinal, then $H(\kappa) \vDash Z - \mathcal{P} - \mathcal{I}$. Furtheremore, $H(\kappa) \vDash \mathcal{F}$, when κ is a regular cardinal.

Proof. 3. Clear, since if $y \subseteq z$, then $\mathrm{trcl}(y) \subseteq \mathrm{trcl}(z)$, so we have a bound on cardinality.

4. Also clear since

$$\mathrm{trcl}\Big(\bigcup \mathcal{F}\Big) = \bigcup_{n \geq 0} \Big(\bigcup^n \bigcup \mathcal{F}\Big) = \bigcup_{n \geq 1} \bigcup^n \mathcal{F} \subseteq \bigcup_{n \geq 0} \bigcup^n \mathcal{F} = \mathrm{trcl}(\mathcal{F})$$

(5. for κ regular): Let $z = \mathrm{dom}(f) \in H(\kappa)$, since $\mathrm{Im}(f) \subseteq H(\kappa)$ and f is a surjection from z to its image, then $|\mathrm{Im}(f)| \leq |z| < \kappa$ which, by Lemma 1.3 implies that $\mathrm{Im}(f) \in H(\kappa)$. ■

2.2 Absoluteness and Power Set Axiom

We shall now consider the remaining three Axioms. As we saw, the previous proofs displayed directly the axioms as relativized $\in$-formulas, which is doable, but complicated for sentences about cardinals and other gigantic definitions. For this reason we will talk about absoluteness.

Let's start proving that every Δ_0 formula is absolute

Lemma 2.2

Let $\in$ be a symbol of $\mathcal{L}$ and $\mathfrak{A} \subseteq \mathfrak{B}$. If A is transitive, then $\mathfrak{A} \preceq_\varphi \mathfrak{B}$ for all Δ_0 $\mathcal{L}$-formulas φ. For M transitive and $B = V$ it says we can swap from φ^M to φ.

Proof. The base case for φ is clear since quantifier-free formulas are preserved under substructures. Similarly, the induction steps for the logical connectives are also clear.

For $\exists$, let $\varphi(\overline{v}) = \exists y[y \in \tau(\overline{v}) \wedge \psi(\overline{v}, y)]$, where ψ is Δ_0 and $\mathfrak{A} \preceq_\psi \mathfrak{B}$ as the induction hypothesis. Then, let $\overline{a} \in A^n$, we claim that we have the following:

$$\begin{aligned} \mathfrak{A} \models \varphi(\overline{a}) &\Leftrightarrow \exists b \in A \text{ such that } b \in \tau^{\mathfrak{A}}(\overline{a}) \text{ and } \mathfrak{A} \models \psi(\overline{a}, b) \\ &\Leftrightarrow \exists b \in B \text{ such that } b \in \tau^{\mathfrak{B}}(\overline{a}) \text{ and } \mathfrak{B} \models \psi(\overline{a}, b) \Leftrightarrow \mathfrak{B} \models \varphi(\overline{a}) \end{aligned}$$

The ($\Rightarrow$) direction is clear, since $\tau^{\mathfrak{A}}(\overline{a}) = \tau^{\mathfrak{B}}(\overline{a})$ and $A \subseteq B$ and $\mathfrak{A} \preceq_\psi \mathfrak{B}$. For the ($\Leftarrow$) direction, since A is transitive and $\tau^{\mathfrak{A}}(\overline{a}) \in A$, then $\tau^{\mathfrak{A}}(\overline{a}) \subseteq A \subseteq B$. ∎

Example: For some simple examples, note that the following are Δ_0 formulas:

$$\begin{aligned} x \subseteq y &\equiv \forall z \in x(z \in y) \\ y = S(x) &\equiv \forall z \in y(y = x \vee y \in x) \wedge x \in y \wedge x \subseteq y \\ x = \emptyset &\equiv \forall z \in x(z \neq z) \\ y = v \cap w &\equiv \forall z \in y(z \in v \wedge z \in w) \wedge y \subseteq v \wedge y \subseteq w \\ \text{“}x \text{ is a singleton”} &\equiv \exists z \in x\, \forall y \in x(z = y) \end{aligned}$$

We will often denote “x is a singleton” as $\mathrm{Sing}(x)$.

Observation: It's worth noting that, in the previous example, the equivalences are not the “oficial” definitions of those terms, in fact, they are equivalent over some basic set theory. What we are using here is the fact that if $V, M \models \forall \overline{v}(\varphi(\overline{v}) \leftrightarrow \psi(\overline{v}))$, then φ is absolute for M iff ψ is also absolute for M. What the previous statements says is that M should be able to prove the logical equivalence between the formulas and, then, we sometimes require M to satisfy some basic axioms of ZFC.

With absoluteness in hands, we can prove the following lemma

Lemma 2.3

Let M be a transitive class, then

$$\forall x \in M((\mathcal{P}(x) \cap M) \in M) \Rightarrow M \vDash \mathcal{P}$$

Also, we have the ($\Leftarrow$) direction if $M \vDash \mathcal{C}$.

Proof. By absoluteness of $\subseteq$, the Power Set Axiom relativized to M is equivalent to

$$\forall x \in M\ \exists y \in M\ \forall z \in M(z \subseteq x \to z \in y)$$

Then it's easy to see, for each $x \in M$, we can just take $y = \mathcal{P}(x) \cap M$ which is in M by hypothesis. The other direction is also clear, since $M \vDash \mathcal{P}$, then for each $x \in M$ there is some $y \in M$ such that $\mathcal{P}(x) \cap M \subseteq y$. By Comprehension and absoluteness of $\subseteq$ we have that

$$\mathcal{P}(x) \cap M = \{z \in y : (z \subseteq x)^M\} \in M$$

■

The following is clear by the definition of R and since $WF = R(ON)$.

Corollary 2.4

$WF \vDash \mathcal{P}$ and $R(\gamma) \vDash \mathcal{P}$ for any limit γ.

Unfortunately (or not since it makes independence easy) the same is not true for $H(\kappa)$, as we will see in subsubsection 2.4.1

2.3 Axiom of Infinity and Choice

As we shall see, the Axiom of Infinity and AC can be stated only using symbols we proved in the last subsection to be absolute, then we don't need to worry about their relativizations.

Lemma 2.4

Let M be a transitive class such that $M \vDash \mathcal{E}, \mathcal{U}, \mathcal{C}$ and Pairing.

1. $M \vDash \mathcal{I}$ iff $\omega \in M$;
2. $M \vDash$ AC iff every disjoint family of non-empty sets in M has a choice set in M.

Proof. 1. ($\Rightarrow$) The Axiom of Infinity is equivalent to $\exists x$ Ind(x), where Ind(x) states that x is inductive, i.e.

$$\text{Ind}(x) \equiv \emptyset \in x \wedge \forall y \in x(S(y) \in x)$$

Since it's absolute, then $\mathcal{I}^M = \exists x \in M(\text{Ind}(x))$, whose $x = \omega$ clearly satisfies.

($\Leftarrow$) Since Ind is absolute, given that there is some $\lambda \in M$ such that Ind(λ), then λ is in fact a limit ordinal and, by transitiveness of M, all smaller ordinals are also in M, in particular, $\omega \in M$.

2. AC is equivalent to $\forall F\ \exists C[\mathrm{df}(F) \to \mathrm{cs}(C, F)]$, where $\mathrm{df}(F)$ states F is a disjoint family of non-empty sets and $\mathrm{cs}(C, F)$ that C is a choice set for F, i.e.

$$\mathrm{df}(F) \equiv \emptyset \notin F \wedge \forall x \in F\ \forall y \in F(x \neq y \to x \cap y = \emptyset)$$
$$\mathrm{cs}(C, F) \equiv \forall x \in F(\mathrm{Sing}(C \cap x))$$

which are all absolute. Then $\mathrm{AC}^M = \forall F \in M\ \exists C \in M[\mathrm{df}(F) \to \mathrm{cs}(C, F)]$. which is our condition. ■

The hypothesis that M should satisfy $\mathcal{E}$, $\mathcal{U}$, $\mathcal{C}$ and Pairing was used in the absoluteness of the Δ_0 formulas, due to the observation after Lemma 2.2.

Observation: As a fun fact, we can't prove the ($\Rightarrow$) direction of (1) without Regularity. In order to show this we will need to analyze permutations in V and use Mostowski Collapsing Lemma, which we will see in the following sections.

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We can now analyze if these axioms are satisfied by our approximate models. In fact, we will see in subsection 2.4.5 that $WF \models \mathrm{ZFC}$. For now, let's stick to $R(\gamma)$ and $H(\kappa)$.

Corollary 2.5

$R(\gamma) \models ZC$ whenever $\gamma > \omega$ is a limit ordinal.

Proof. Since $\gamma > \omega$, clearly $\omega \in R(\gamma)$. Now, given $F \in R(\gamma)$ a family of disjoint nonempty sets, by AC there is some choice set C, choose it such that $C \subseteq F$ (you can take, for example, $C \cap \bigcup F$). Since $C \subseteq F \in R(\gamma)$ and γ is a limit ordinal, then $C \in P(F) \subseteq R(\gamma)$.

Then, by Corollary 2.1 and 2.2 we have that $R(\gamma) \models ZC$. ■

Corollary 2.6

$H(\kappa) \models \mathrm{ZFC} - \mathcal{P}$ whenever κ is a regular uncountable cardinal.

Proof. Since $\kappa > \omega$, clearly $\omega \in H(\kappa)$. Now, given $F \in H(\kappa)$ a family of disjoint nonempty sets, we can take $C \subseteq F$ a choice set as in the previous proof. Since $F \in H(\kappa)$ and $H(\kappa)$ is transitive, then $F \subseteq H(\kappa)$ and, therefore, $C \subseteq H(\kappa)$. Now, since κ is regular and $|C| \leq |F| < \kappa$, then, by Lemma 1.3, $C \in H(\kappa)$, as desired.

The rest follows from Corollary 2.1. ■

2.4 Independence of ZFC Axioms

We will now use our approximate models to construct models of $\mathrm{ZFC} - \varphi$, where φ is some axiom (or axiom schema) of ZFC, with the exception of the basic axioms and AC, whose independence is way harder and will be shown using L and permutations of V in the following sections.

2.4.1 Power Set

Let's analyze when Power Set Axiom is true in $H(\kappa)$.

Lemma 2.5

If κ is regular, uncountable and not strongly inaccessible, then $H(\kappa) \vDash \neg\mathcal{P}$.

Proof. Let's use Lemma 2.2. Since κ is not strongly inaccessible, there is some $\lambda < \kappa$ such that $2^\lambda \geq \kappa$. And, since κ is also regular, by Lemma 1.3 we have that $\mathcal{P}(\lambda) \subseteq H(\kappa)$. However, note that

$$|\mathrm{trcl}(\mathcal{P}(\lambda))| \geq |\mathcal{P}(\lambda)| = 2^\lambda \geq \kappa$$

and then, $\mathcal{P}(\lambda) \cap H(\kappa) = \mathcal{P}(\lambda) \notin H(\kappa)$, which implies $H(\kappa) \vDash \neg\mathcal{P}$, since $H(\kappa) \vDash \mathcal{C}$. ■

Observation: Furthermore, when κ is strongly inaccessible, we have $H(\kappa) \vDash \mathcal{P}$. To see this, note first that, as a scholium of Lemma 1.3, we have that, for each $x \in H(\kappa)$, with $|x| = \lambda$

$$\mathrm{trcl}(\mathcal{P}(x)) = \mathcal{P}(x) \cup \bigcup_{y \in \mathcal{P}(x)} \mathrm{trcl}(y)$$

and, therefore, knowing that $y \subseteq x$ implies $\mathrm{trcl}(y) \subseteq \mathrm{trcl}(x)$, we also have

$$\begin{aligned} |\mathrm{trcl}(\mathcal{P}(x))| &\leq |\mathcal{P}(x)| + \sum_{y \in \mathcal{P}(x)} |\mathrm{trcl}(y)| \\ &\leq |\mathcal{P}(x)| + |\mathcal{P}(x)||\mathrm{trcl}(x)| \\ &= 2^\lambda + 2^\lambda |\mathrm{trcl}(x)| \\ &< \kappa^2 = \kappa && (2^\lambda < \kappa) \end{aligned}$$

then $\mathcal{P}(x) \in H(\kappa)$ as desired.

We can, therefore, prove our first independency result for a nontrivial axiom.

Theorem 2.1

The Axiom of Power Set is independent of ZFC $- \mathcal{P}$.

Proof. Consider $HC = H(\aleph_1)$. Since $\aleph_1$ is a regular uncountable cardinal, but not strongly inaccessible, the result follows trivially from Corollary 2.3 and the previous lemma. ■

Observation: It can be argued that it would be way easier (but not that cool) to just show HC proves $\mathcal{P}(\omega)$ doesn't exists. However it would require some absoluteness results from the next section, which is really boring to prove. Let's sketch how the proof would be:

Proof. Clearly $\forall x \in HC\ \exists f \in HC\ [f : x \overset{1-1}{\to} \omega]$, since every set in HC is countable. Now, since $f : x \overset{1-1}{\to} \omega$ is absolute, then we can't have $\mathcal{P}(\omega) \in HF$. ■

Way shorter, right? But trust me, the details are boring.

2.4.2 Infinity

Theorem 2.2

The Axiom of Infinity is independent of $\mathrm{ZFC} - \mathcal{I}$

Proof. Consider $HF = H(\aleph_0) = R(\omega)$. Since $\omega \notin R(\omega)$, by Lemma 2.3 $HF \vDash \neg\mathcal{I}$. Also, since HF can be well-ordered, then a choice set always exists in it, then $HF \vDash \mathrm{AC}$.

The other axioms are consequences of Corollaries 2.1 and 2.2. ■

2.4.3 Replacement

Before our proof, let's prove the following lemma about Beth function:

Lemma 2.6

Show that $\{\delta : \delta = \beth_\delta\}$ is unbounded in ON.

Proof. Define δ_ξ by transfinite recursion as follows:

- $\delta_0 = 0;\qquad \delta_{\xi+1} = \beth_{\delta_\xi};$
- $\delta_\eta = \sup\{\delta_\xi : \xi < \eta\}$, when η is a limit ordinal.

Note first that, given $\alpha < \delta_\eta$, with η a limit ordinal, since $\delta_\eta = \sup\{\delta_\xi : \xi < \eta\}$, there is some $\xi < \eta$ such that $\delta_\xi > \alpha$. Using this fact, we then have that, for η a limit ordinal

$$\beth_{\delta_\eta} = \sup\{\beth_\alpha : \alpha < \delta_\eta\} = \sup\{\beth_{\delta_\xi} : \xi < \eta\} = \sup\{\delta_{\xi+1} : \xi < \eta\} = \delta_\eta$$

Since η is an arbitrary limit ordinal, we conclude that $\{\delta : \delta = \beth_\delta\}$ is unbounded in ON. ■

Let's now see a necessary condition for $R(\gamma) \vDash \mathcal{F}$.

Lemma 2.7

If $R(\gamma) \vDash \mathcal{F}$, for $\gamma > \omega$, then $\gamma = \beth_\gamma$ and $\{\delta < \gamma : \delta = \beth_\delta\}$ has size γ.

Proof. PENDENT ■

It should be a direct corollary of the previous lemma using that $\delta \mapsto \beth_\delta$ is absolute

Theorem 2.3

The Axiom of Replacement is independent of ZC

Proof. Consider $M = R(\omega + \omega)$, by Corollary 2.3 it's a model of ZC and, by the previous lemma, since $\omega + \omega \neq \beth_{\omega+\omega}$, we have that $M \vDash \neg\mathcal{F}$. ■

2.4.4 Union

This is not as direct as the previous theorems, in fact we need to define some new kind of model approximation of V that looks like $H(\kappa)$.

Definition 2.1

Given a cardinal κ, a set x is pseudo-hereditarily of cardinality less than κ if $|x| < \kappa$ and, for all $y \in x$, $|\mathrm{trcl}(y)| < \kappa$, i.e., $y \subseteq H(\kappa)$. We can define

$$H^*(\kappa) := \{x : x \text{ is pseudo-hereditarily of cardinality less than } \kappa\}$$

Intuitively, this is as $H(\kappa)$, but it doesn't restrict the size of the union $x \cup \bigcup_{y \in x} \mathrm{trcl}(y)$, just the size of each term. By the next lemmas, our intended model for $\mathrm{ZFC} - \mathcal{U} + \neg\mathcal{U}$ will be $H^*(\beth_\omega)$.

Lemma 2.8

Let κ be a strong limit cardinal, then $H^*(\kappa) \vDash \mathrm{ZFC} - \mathcal{U}$

Proof. Clearly $H(\kappa) \subseteq H^*(\kappa) \subseteq WF$ for all cardinals κ. And, if $x \in y \in H^*(\kappa)$, then $x \subseteq \mathrm{trcl}(y)$, which implies that $|x| \leq |\mathrm{trcl}(y)| < \kappa$. Similarly, for all $z \in \mathrm{trcl}(x) \subseteq \mathrm{trcl}(y)$, we have $\mathrm{trcl}(z) \subseteq \mathrm{trcl}(y)$, then $|\mathrm{trcl}(z)| \leq |\mathrm{trcl}(y)| < \kappa$ and, therefore, $x \in H^*(\kappa)$.

Since $H^*(\kappa)$ is a subset of WF and transitive, by Lemma 2.1, it satisfies $\mathcal{E}$ and $\mathcal{R}$. Also, since $\kappa > \aleph_0$, we have $\omega \in H^*(\kappa)$ and, by Lemma 2.3, it satisfies $\mathcal{I}$.

For Power Set Axiom, we will use Lemma 2.2. If $x \in H^*(\kappa)$ and $y \subseteq x$, then $|y| \leq |x| < \kappa$ and $\mathrm{trcl}(y) \subseteq \mathrm{trcl}(x)$, so $|\mathrm{trcl}(y)| \leq |\mathrm{trcl}(x)| < \kappa$, this implies that $\mathcal{P}(x) \cap H^*(\kappa) = \mathcal{P}(x)$. Now, if $|x| = \lambda < \kappa$, since κ is strong limit, then $|\mathcal{P}(x)| = 2^\lambda < \kappa$. And, for $y \in \mathcal{P}(x)$, as we saw $|\mathrm{trcl}(y)| < \kappa$, which implies that $\mathcal{P}(x) \in H^*(\kappa)$, then $H^*(\kappa)$ satisfies $\mathcal{P}$.

As in the proof that $H(\kappa) \vDash \mathcal{F}$ and $\mathcal{C}$, we just need to show the analogous of the Lemma 1.3 for $H^*(\kappa)$. Given $x \subseteq H^*(\kappa)$ such that $|x| < \kappa$, if $y \in \mathrm{trcl}(x)$, then $\mathrm{trcl}(y) \subseteq \mathrm{trcl}(x)$, which implies that $|\mathrm{trcl}(y)| \leq |\mathrm{trcl}(x)| < \kappa$, therefore $x \in H^*(\kappa)$ and $H^*(\kappa) \vDash \mathcal{F}, \mathcal{C}$. ■

Lemma 2.9

If κ is a singular strong limit cardinal, then $H^*(\kappa) \vDash \neg\mathcal{U}$

Proof. Let $\mathrm{cf}(\kappa) = \lambda$ and $\langle \alpha_\beta : \beta < \lambda \rangle$ be a strictly increasing sequence converging to κ, with $|\alpha_\beta| < \kappa$, for every $\beta < \lambda$. Clearly $\xi \in H^*(\kappa)$ for every ξ such that $|\xi| < \kappa$, then $\lambda, \alpha_\beta \in H^*(\kappa)$, for every $\beta < \lambda$, and, therefore, $\{\alpha_\beta : \beta < \lambda\} \in H^*(\kappa)$ since it satisfies $\mathcal{F}$. However

$$\bigcup \{\alpha_\beta : \beta < \lambda\} = \kappa \notin H^*(\kappa)$$

then $H^*(\kappa) \vDash \neg\mathcal{U}$ ■

Observation: It's clear that, in the previous model, The Axiom of Union fails to ensure that the union of a set of size $\mathrm{cf}(\kappa)$ exists, in particular, the model of the next theorem doesn't guarantee that union of countable sets always exists.
Fortunately, since $\beth_\alpha$ is strong inaccessible, whenever α is a limit ordinal, and since $\mathrm{cf}(\kappa)$ is always a regular cardinal, then $H^*(\beth_{\kappa+\kappa})$, with κ regular, is a model where The Axiom of Union fails for sets of size κ, since $\beth$ is a continuous function, $\mathrm{cf}(\beth_{\kappa+\kappa}) = \kappa$ and $\beth_{\kappa+\kappa} > \beth_\kappa \geq \mathrm{cf}(\beth_\kappa) = \kappa$, i.e., it's singular.
Some natural questions are whether we can find models whose union fails for singular cardinals or, maybe, if we can make it fails even for union of finite elements.
It turns out that, surprisingly, we can make it fail for the union of two elements! The full construction can be found here, unfortunately, it uses forcing, which we didn't learn yet.

Theorem 2.4

The Axiom of Union is independent of $\mathrm{ZFC} - \mathcal{U}$.

Proof. By the two previous lemmas, let $M = H^*(\beth_\omega)$, since $\beth_\omega$ is a singular strong limit cardinal, then $M \vDash \mathrm{ZFC} - \mathcal{U} + \neg\mathcal{U}$, as desired. ■

2.4.5 Regularity

This is way easier than the previous result, but the results in this subsubsection are in some sense more important, that's why it's the last one.

Theorem 2.5

The Axiom of Regularity is independent of ZFC^-.

Proof. Let's add the constant symbols $\{c_n\}_{n\in\omega}$ to the language. Consider now the set

$$S = \{c_{n+1} \in c_n : n \in \omega\}$$

Now, since we're assuming ZFC is consistent, let M be some model of it. Clearly for any finite subset $S_0 \subseteq S$ we have that $M \vDash S_0$, then, by Compactness Theorem, there is some model M' of $\mathrm{ZFC} + S$, which gives us an infinite descending $\in$-chain $c_0 \ni c_1 \ni c_2 \ni \ldots$, therefore $M' \vDash \neg\mathcal{R}$. ■

It's worth noting that we can also construct some model of ZF^- that doesn't satisfies regularity using permutations of V, as we shall see in the following sections.

We have two reasons to give a different proof. The first is because the method used introduces a really powerful method called Permutation Models. The second is because it can be hard to tell how many axioms of ZFC we need to prove Compacteness. In fact, we need at least AC, since Compacteness is equivalent to the Ultrafilter Lemma, which is a weaker fragment of AC. With the alternative proof, however, we can just work with ZF^-, instead.

Let's conclude this section finishing the proof that WF is a model of ZFC.

Theorem 2.6

$WF \vDash \mathrm{ZF}$ and, if we assume AC, $WF \vDash \mathrm{ZFC}$.

Proof. It follows easily from Corollaries 2.1 and 2.3 with the fact that $WF = R(ON)$. ■

It's worth noting now that, almost all work we did from now on can be done in $\text{ZF}^- - \mathcal{P}$ instead of ZFC, which means we could prove the following results about proof theoretic equivalence

Theorem 2.7

Let Γ be one of the theories $\text{ZF} - \mathcal{P}$, $\text{ZFC} - \mathcal{P}$, ZF, ZFC. If $\Gamma^- = \Gamma - \mathcal{R}$, then $\Gamma \leq \Gamma^-$.

Proof. Working in Γ^-, we can just prove, as before, each axiom of Γ relativized to WF. ■

And, since $\Gamma^- \leq \Gamma$, then we conclude that $\Gamma \sim \Gamma^-$ which, in some sense, says that set theory without regularity is not weaker, i.e., regularity don't really add anything new.

Note also that, since in almost all other independence proofs we used models that are actually sets, then we proved the stronger claim that $\Gamma \lhd \text{ZFC}$, where Γ is ZFC minus that specific axiom. This implies that, by Lemma 1.1, with the exception of regularity, Γ can't be proof theoretically equivalent to ZFC. We will see later that GHC is also another exception.

With this remarks we will, for now on, always assume the Regularity Axiom (which will help us).

CHAPTER 3

The Constructible Universe L

3.1 About Inner Model Theory

From here on, we will construct some kind of "Inner Models" of Set Theory, namely L and HOD. In fact, such models assume we already have a model M of ZFC, then we actually should denote them by L^M and HOD^M.

We can give an intuitive idea on how the method of inner models work using a simpler example. Let Φ be the axioms of a group, we already know that the property of being abelian, i.e., $\varphi = \forall x, y(x \cdot y = y \cdot x)$ is independent of Φ, since there are a lot of abelian and non abelian groups. In fact, we can show φ is consistent with Φ by exhibiting an inner model: Assume $\mathrm{Con}(\Phi)$, then there is a $G \models \Phi$ and, in particular, we know $Z^G \models \Phi + \varphi$, where Z^G is the center of G.

For the constructible universe L, a better analogy would be to consider the inner model $E^G = \{e_G\}$, where we only have the identity element of G. Clearly E^G is the minimal inner model, just as L^M.

The only difference from set theory is that we can't actually show that $\mathrm{Con}(\Phi) \to \mathrm{Con}(\Phi + \varphi)$, since the theory of groups is way weaker than ZFC.

3.2 More on Absoluteness

I will complete it when needed

This subsection, although filled with a lot of boring proofs, will be important in general to avoid future stress. In particular, we will prove a useful criterion to verify when a transitive set is a model for ZF, which will simplify our future proofs for L and HOD.

As a first example of the importance of Regularity Axiom, we have the following Lemma who shows that notions like "ordinal", and related concepts, now become absolute over transitive models.

Lemma 3.1

The following notions are equivalent to formulas BST proves are equivalent to Δ_0 formulas and, therefore, absolute over transitive models of BST:

1. x is transitive;
2. x is an ordinal;
3. x is a successor ordinal;
4. x is a finite ordinal;
5. x is a limit ordinal;
6. $x = \omega$.

It should also be useful to remember the observation of subsection 2.2 about absoluteness.

Proof. 1. x is transitive iff $\forall y \in x(\forall z \in y(z \in x))$;

2. x is an ordinal iff x is transitive (which we already saw is absolute) and it's well ordered by $\in$. If we look at the definition of well ordering we will see every condition is relativized to x itself, since it's all about elements of it, then it's already a Δ_0 formula;

3. x is a successor ordinal iff it's an ordinal (absolute) and $x = S(y)$ for some y also an ordinal. But we saw in subsection 2.2 that $x = S(y)$ is equivalent to a Δ_0 formula;

4. x is a finite ordinal iff x is an ordinal and $\forall y \in x(y = 0 \vee$ "y is a successor ordinal");

5. x is a limit ordinal iff it's an ordinal that is not successor;

6. $x = \omega$ iff x is a limit ordinal and $\forall y \in x($"y is a successor ordinal"$)$. ■

Let's now use it to prove that some other concepts are also absolute.

Lemma 3.2

The following are absolute over transitive models of BST:

1. The 2-ary function $\cup$ and the 1-ary function $\bigcup$;
2. The ternary relation $\{x, y\} = z$;
3. The 2-ary function $x, y \mapsto \{x, y\}$;
4. The properties: z is an ordered pair, and x is a relation;
5. $\mathrm{dom}(x)$ and $\mathrm{Im}(x)$;
6. The properties: f is a function, injectiveness and surjectiveness;
7. The function $x, y \mapsto x \times y$;
8. R being transitive, reflexive, irreflexive, symmetric and satisfying trichotomy.

Proof. 1. $y = a \cup b$ if, and only if, $\forall x \in y(x \in a \vee x \in b)$, similarly we have

$$y = \bigcup x \iff \forall z \in x(z \subseteq y) \wedge \forall w \in y\ \exists v \in x(w \in v);$$

2. and 3. $z = \{x, y\}$ if, and only if, $\forall w \in z(z = x \vee z = y) \wedge x \in z \wedge y \in z$, but BST also proves that for all x and y, there is a unique z such that $z = \{x, y\}$, then we can define such function;

4. By the previous item, $z = (x, y)$ is also absolute and, therefore, z is an ordered pair if, and only if, $\exists w \in z\, \exists x, y \in w(z = (x, y))$. Then, x is a relation iff $\forall y \in x$("y is an ordered pair").

5. We have that $y = \mathrm{dom}(x)$ if, and only if

$$\forall z \in y\, \exists w \in \bigcup\bigcup x\, [(z, w) \in x] \wedge \forall z, w \in \bigcup\bigcup x\, [(z, w) \in x \to z \in y]$$

Similarly for $y = \mathrm{Im}(x)$.

6. Using the previous items, f is a function if, and only if

$$\forall x \in \mathrm{dom}(f)\, \exists! y \in \mathrm{Im}(f)\, [(x, y) \in f] \equiv \forall x \in \mathrm{dom}(f)\, \forall y, z \in \mathrm{Im}(f)\, [(x, y) \in f \wedge (x, z) \in f \to y = z]$$

also, f is injective iff $\forall x, y \in \mathrm{dom}(f)\ [f(x) = f(y) \to x = y]$, surjectiveness is similar.

7. We have that $z = x \times y$ if, and only if

$$\forall a \in x\, \forall b \in y\, [(a, b) \in z] \wedge \forall w \in z\, \exists a \in x\, \exists b \in y\, [w = (a, b)]$$

8. This is easy to see, since all these definitions quantify over $R \subseteq A^2$ or A. ∎

Absoluteness of Ordinals Operations and Relations

We can finally show "finite" and "hereditarily finite" are absolute over transitive models of BST.

Lemma 3.3

Let M be a transitive model of BST, then

1. $[M]^{<\omega} \subseteq M$;
2. $HF \subseteq M$;
3. $M^{<\omega} \subseteq M$.

Proof. 1. We will prove it by (metatheorical) induction over ω. Clearly $[M]^0 = \{\emptyset\} \subseteq M$. Now, assume $[M]^n \subseteq M$, clearly every $x \in [M]^{n+1}$ can be written as a union $y \cup \{a\}$, where $a \notin y \in [M]^n$ and $a \in M$. In fact, by Lemma 3.2, the functions $x, y \mapsto \{x, y\}$ and $a, b \mapsto a \cup b$ are absolute. In particular, given $a \in M$ and $y \in [M]^n$, we have $\{a, a\} = \{a\} \in M$ and $y \cup \{a\} \in M$.

2. Let's also use induction over ω. Trivially $R(0) = \emptyset \subseteq M$. Now, if $R(n) \subseteq M$, then, given $x \in R(n+1) = \mathcal{P}(R(n))$, we have $x \subseteq R(n) \subseteq M$, then $x \subseteq M$ and, since $R(n)$ is finite, for every $n \in \omega$, then x is also finite and, therefore, $x \in [M]^{<\omega} \subseteq M$, then, by transitivity, $x \in M$.

3. Again, by induction, the proof is similar to (1). If $f : 0 \to M$, then $f = \emptyset \in M$. Now, given that every $g : n \to M$ is in M, let $f : n + 1 \to M$. Clearly $f = (f \restriction_n) \cup \{(n, f(n))\}$, where $f \restriction_n : n \to M$ is in M by hypothesis. Now, $f(n) \in M$, since $f : n + 1 \to M$, and $n \in M$, since by Lemma 3.2, being natural is absolute and, by (2), $HF \subseteq M$. We then have that $\{(n, f(n))\} \in M$ by the same lemma we used in (1), and also f itself, since it's the union of both. ∎

The following is the criterion we talked about in the start of this subsection. In fact it's fairly easy to prove that, under such hypothesis, we have that $M \vDash \mathrm{ZF} - \mathcal{I}$, the problem is for infinity, where now we have a really nice proof using absoluteness of HF over transitive models of BST.

Lemma 3.4

Let M be a transitive class satisfying Comprehension. If for every $x \subseteq M$, there is a $y \in M$ such that $x \subseteq y$, then $M \vDash \text{ZF}$.

Proof. $M \vDash \mathcal{C}$ by hypothesis and $M \vDash \mathcal{R}$ by regularity. Now, by Lemma 2.1: since M is transitive, $M \vDash \mathcal{E}$. Also, given $\mathcal{F} \in M$, then $x \in \bigcup \mathcal{F}$ iff $x \in y$ for some $y \in \mathcal{F} \in M$, by transitivity of M we have $x \in M$, then $\bigcup \mathcal{F} \subseteq M$ and, by hypothesis, there is some $y \in M$ such that $\bigcup \mathcal{F} \subseteq y$ and, by the Comprehension Axiom we can conclude that

$$\bigcup \mathcal{F} = \{x \in y : \exists z \in \mathcal{F}\ (x \in z)\} \in M$$

And, for Substitution, let $f : z \to M$, with $z \in M$. Since $\text{Im}(f) \subseteq M$, then there is some $y \in M$ such that $\text{Im}(f) \subseteq y$ and, also by the Comprehension Axiom:

$$\text{Im}(f) = \{x \in y : \exists w \in z\ (x = f(z))\} \in M$$

Now, by Lemma 2.2, clearly $M \vDash \mathcal{P}$.

Finally, by Lemma 3.2, since $M \vDash \text{BST}$ and it's transitive, then $HF \subseteq M$. In particular, $\omega \subseteq M$, then there is some $y \in M$ such that $\omega \subseteq y$ and, again by Comprehension

$$\omega = \{n \in y : n \text{ is finite}\} \in M$$

by absoluteness of finiteness and by Lemma 2.3 we have that $M \vDash \mathcal{I}$. ■

The following Theorem will be crucial in the proof that $(V = L)^L$, it basically guarantees us that, if we define some function F with domain A recursively using the constructor G, and both A and G are absolute, then F is also. We will state the more general statement using any R well founded.

Theorem 3.1

Let R be a well founded relation on A and F be defined by recursion on A using G, i.e., where $\forall a \in A\,[F(a) = G(a, F\restriction_{\text{pred}(a)})]$ and $F(a) = \emptyset$ for $a \notin A$. If $M \vDash \text{ZF} - \mathcal{P}$ is transitive, R, A and G are absolute over M, and $\text{pred}(a) \subseteq M$ whenever $a \in M$, then $F^M(a)$ is defined for all $a \in M$, and F is absolute for M.

Proof. Note that $(R \text{ is well founded on } A)^M$, since any $\emptyset \neq X \in M$ with $X \subseteq A$ that doesn't have a R-minimal element would contradict the well foundedness of R. Also, $a \mapsto \text{pred}(a)$ is absolute for M, since $x = \text{pred}(a)$ iff $\forall y \in x(yRa) \wedge \forall b \in A(bRa \to b \in x)$. Now, using the Recursion Theorem, F^M is defined and, by absoluteness of G, if $F\restriction_{\text{pred}(a)} = F^M\restriction_{\text{pred}(a)}$, then $F(a) = F^M(a)$.

Finally, if F were not absolute, we could find a R-minimal element a of $\{a \in M : F(a) \neq F^M(a)\}$, but then $F(b) = F^M(b)$ for every bRa and, then, $F\restriction_{\text{pred}(a)} = F^M\restriction_{\text{pred}(a)}$, a contradiction. ■

3.3 Mostowski Collapse Lemma

For everything we saw until now, transitive sets are really well behaved and it would be really cool if we could always work with a transitive model. This subsection will be devoted exactly to the proof of the following more general fact, called Mostowski Collapse Lemma:

Lemma 3.5

(Mostowski Collapse) Let $(B, \in)$ be a model of Extensionality and let $\kappa \leq |B|$ be any infinite cardinal. Fix $S \subseteq B$ such that S is transitive and $|S| \leq \kappa$. Then there is a transitive M such that $S \subseteq M$, $(M, \in) \equiv (B, \in)$ and $|M| = \kappa$.

This is also called transitive collapse, and, as we will see in the proof, is a stronger corollary to the fact that for any model $(B, \in)$ of Extensionality, there are unique M and π such that $\pi : (B, \in) \cong (M, \in)$. Before giving its proof, let's motivate with the following sketch:

Since $\in$ is metatheoretically well-founded and extensional in B, we can look at the $\in$-minimal elements b_0 of B and send them to 0. Then we take the next $\in$-smallest sets b_1 and send to $1 = \{0\}$. The following definition capture formally this idea using transfinite recursion:

Definition 3.1

Given a well-founded relation R on A define recursively, for $y \in A$,

$$\mathrm{mos}_R(y) = \{\mathrm{mos}_R(x) : xRy\}$$

For our specific case, $R = \in$ and $\mathrm{mos}(y) = \{\mathrm{mos}(x) : x \in y\}$, this will be our unique isomorphism π. Technically we should write $x \in A$ and xRy in the definition of $\mathrm{mos}_R(y)$, but it should be clear that this is the only way it would make sense.

Clearly by definition $\mathrm{mos}_R[A]$ is always a transitive set. The following lemma justifies the additional assumption that B should satisfy Extensionality Axiom.

Lemma 3.6

If R is well-founded on A, then mos_R is injective iff (A, R) satisfies Extensionality, in which case $\mathrm{mos} : (A, R) \cong (\mathrm{mos}[A], \in)$, i.e., it gives us an isomorphism.

Proof. $(\Rightarrow)$ If (A, R) doesn't satisfy Extensionality, then there are $a, b \in A$, with $a \neq b$, such that $\forall x(xRa \leftrightarrow xRb)$, then $\mathrm{mos}(a) = \mathrm{mos}(b)$ and it's not injective.

$(\Leftarrow)$ Assume for the sake of a contradiction that (A, R) satisfy Extensionality and mos isn't injective. Then there is an R-least element $a \in A$ such that there is some $b \neq a$ satisfying $\mathrm{mos}(b) = \mathrm{mos}(a)$. Since $(A, R) \vDash \mathcal{E}$ and $a \neq b$ we have $\mathrm{pred}_R(a) \neq \mathrm{pred}_R(b)$, then there are two cases:

- If there is a c with cRa and $\neg cRb$, then $\mathrm{mos}(c) \in \mathrm{mos}(a) = \mathrm{mos}(b) = \{\mathrm{mos}(d) : dRb\}$ implying that $\mathrm{mos}(c) = \mathrm{mos}(d)$ for some dRb. Since $\neg cRb$, then $c \neq d$, but such c contradicts the minimality of a;
- If there is a d with $\neg dRa$ and dRb, then we can repeat the exact same argument and find a c that contradicts the minimality of a.

What finishes the proof. ∎

And finally, the following is the last Lemma we will need to prove Mostowski Collapse

Lemma 3.7

Let $T \subseteq A$ be transitive. If $(A, \in) \vDash \mathcal{E}$, then $\mathrm{mos}(y) = y$ for all $y \in T$.

Proof. Assume that $\text{mos}(y) \neq y$ for some $y \in T$. Let a be the $\in$-minimal element of T such that $\text{mos}(a) \neq a$. If $y \in a \in T$, since T is transitive, then $y \in T$, and by minimality of a we have

$$\text{mos}(a) = \{\text{mos}(y) : y \in A \wedge y \in a\} = \{y : y \in a\} = a$$

a contradiction. ■

We are now ready to prove Mostowski Collapse Lemma

Proof. By Downward Löwenheim-Skolem let $A \preceq B$ with $S \subseteq A$ and $|A| = \kappa$, then $(A, \in)$ also satisfies Extensionality. By Lemma 3.3 mos is an isomorphism from $(A, \in)$ onto some transitive set $(M, \in)$. Furthermore, since S is transitive, by the previous Lemma $\text{mos}(y) = y$ for all $y \in S$ and, therefore, $S \subseteq M$. We also have $|M| = \kappa$, since mos is a bijection and, since $M \cong A$ and $A \preceq B$, then $(M, \in) \equiv (A, \in) \equiv (B, \in)$, as desired. ■

It will be used a lot with $S = \emptyset$ and $\kappa = \aleph_0$ to transform models into countable transitive ones.

3.4 Reflection Theorem

3.4.1 Digression About Löwenheim-Skolem

In order to motivate the proof of The Reflection Theorem let's look for the proof of Downward Löwenheim-Skolem Theorem. It's really common to prove Löwenheim-Skolem as a scholium of the Completeness Theorem, since we use a syntactical countable model formed by equivalence classes of some formulas, which automatically gaves us Löwenheim-Skolem. However, there is a different approach we can give using the following lemma, known as Tarski-Vaught Criterion

Lemma 3.8

(Tarski-Vaught) Let $\mathfrak{A} \subseteq \mathfrak{B}$ be $\mathcal{L}$-structures. The following are equivalent

- $\mathfrak{A} \preceq \mathfrak{B}$;
- For all $\varphi \in \mathcal{L}$ and $\overline{a} \in A$, if $\mathfrak{B} \vDash \exists b\ \varphi(\overline{a}, b)$, then there is some $b \in A$ such that $\mathfrak{B} \vDash \varphi(\overline{a}, b)$.

We will omit the proof since we will give a similar one for our specific case and because it's pretty straightforward, we just need to use induction on formulas.

The real power of the previous criterion is that the second statement can be viewed as a closure property of A. In this sense, we can use that to create an elementary substructure of $\mathfrak{B}$.

Let's see how to use it to prove Downward Löwenheim-Skolem Theorem.

Theorem 3.2

(Downward Löwenheim-Skolem) Let $\mathfrak{B}$ be an $\mathcal{L}$-structure. Fix κ such that $\max(|\mathcal{L}|, \aleph_0) \leq \kappa \leq |B|$, and fix $S \subseteq B$ with $|S| \leq \kappa$. Then there is an $\mathfrak{A} \preceq \mathfrak{B}$ such that $S \subseteq A$ and $|A| = \kappa$.

Proof. The idea is to extend S in such a way we can use Tarski-Vaught Criterion. Since the biggest problems are the existential formulas, we extend S in such a way it will be closed under the witnesses. There are two equivalent different ways we can do that, but first, since $S \subseteq B$, $|S| \leq \kappa \leq |B|$, let $S \subseteq S' \subseteq B$ such that $|S'| = \kappa$.

- The first is to define $S' = S_0 \subseteq S_1 \subseteq \cdots \subseteq S_n \subseteq \dots$ such that, for each $\overline{a} \in S_n$ and formula $\varphi(\overline{v}, w)$, if there is some $b \in B$ such that $\mathfrak{B} \vDash \varphi(\overline{a}, b)$ let $f_\varphi(\overline{a}) = b$. Define then

$$S_{n+1} = S_n \cup \{f_\varphi(\overline{a}) : \mathfrak{B} \vDash \exists w\ \varphi(\overline{a}, w),\ \overline{a} \in S_n,\ \varphi \in \mathcal{L}\}$$

  And, finally, $A := \bigcup_{n\in\omega} S_n$. Clearly $|A| = \kappa$.

- The second is to define, for each $\varphi(\overline{v}, w)$, a function $f_\varphi : B^n \to B$, where n depends on φ, and such that for each $\overline{a} \in B^n$, if there is some $b \in B$ satisfying $\mathfrak{B} \vDash \varphi(\overline{a}, b)$, then $f_\varphi(\overline{a}) = b$, otherwise let it be any arbitrary value. We then let $\mathcal{F} = \{f_\varphi : \varphi \in \mathcal{L}\}$ and choose A as the closure of S' under $\mathcal{F}$. It's also not hard to see that $|A| = \kappa$.

It doesn't really matter what approach we take, let's then stick to the first. If c is some constant symbol, then $\mathfrak{B} \vDash \exists w(w = c)$ and $c^{\mathfrak{B}}$ is the only element w of B satisfying $w = c$, then $c^{\mathfrak{B}} = f_{w=c}(\overline{a}) \in S_1 \subseteq A$, for any $\overline{a} \in S_0$. Analogously, given $\overline{a} = a_1, a_2, \dots, a_m \in A$ and an m-ary function symbol f, there is some n such that $a_1, \dots, a_m \in S_n$ and $f^{\mathfrak{B}}(\overline{a})$ is the unique element w of B satisfying $w = f(\overline{a}) = \psi(\overline{a}, w)$, thus $f_\psi(\overline{a}) \in S_{n+1} \subseteq A$.

This guarantees that $\mathfrak{A} \subseteq \mathfrak{B}$, let's now use Tarski-Vaught to show it's an elementary substructure. Given $\varphi(\overline{v}, w) \in \mathcal{L}$ and $\overline{a} \in A$, we know $\overline{a} \in S_n$ for some n. Now, if there is some $b \in B$ such that $\mathfrak{B} \vDash \varphi(\overline{a}, b)$, then $f_\varphi(\overline{a}) \in S_{n+1} \subseteq A$ and $\mathfrak{B} \vDash \varphi(\overline{a}, f_\varphi(\overline{a}))$ by definition and, therefore, $\mathfrak{A} \preceq \mathfrak{B}$. ■

3.4.2 Reflection Theorem and its Consequences

We will replicate the same idea, but for cumulative hierarchies instead of structures in general. The idea is to find such "smaller substructures" that are elementary just for a fixed finite list of formulas. The following is the Tarski-Vaught Criterion for these specific scenarios.

Lemma 3.9

Let $\varphi_1, \varphi_2, \dots, \varphi_n$ be a subformula-closed list of formulas, and let A and B be classes with $\emptyset \neq A \subseteq B$. Then the following are equivalent:

- $A \preceq_{\varphi_i} B$, for $i = 1, \dots, n$;
- For all $\varphi_i(\overline{v}) = \exists w\ \varphi_j(\overline{v}, w)$, the following holds:

$$\forall \overline{a} \in A[\varphi_i^B(\overline{a}) \to \exists b \in A\ \varphi_j^B(\overline{a}, b)]$$

Before proving it, let's explain what "subformula-closed" in the statement means. A list of formulas $\varphi_1, \varphi_2, \dots, \varphi_n$ is subformula-closed iff every subformula of each φ_i is also on the list, and no formula on the list uses the universal quantifier $\forall$. The first condition is to help us in the induction step, since if φ_i is of the form $\exists w\ \varphi_j$, we can use the induction hypothesis on φ_j. And the second condition about the universal quantifier is just to avoid extra work, since we just replace every occurrence of $\forall$ by $\neg\exists\neg$, so it makes no difference.

Proof. The proof is kinda exactly the same as Tarski-Vaught.

($\Rightarrow$) Given $\varphi_i(\overline{v}) = \exists b\ \varphi_j(\overline{v}, b)$ and $\overline{a} \in A$, if $\varphi_i^B(\overline{a})$, since $A \preceq_{\varphi_i} B$, we have $\varphi_i^A(\overline{A})$, i.e., $\exists b \in A\ \varphi_j^A(\overline{a}, b)$, then, by $A \preceq_{\varphi_j} B$, we have $\exists b \in A\ \varphi_j^A(\overline{a}, b)$ as desired.

($\Leftarrow$) We will prove by induction on the length of φ_i. The base case for atomic formulas and the induction steps for the logical connectives are trivial. Now, given $\varphi_i(\overline{v}) = \exists b\ \varphi_j(\overline{v}, b)$, let's fix $\overline{a} \in A$. Then, by hypothesis, $\varphi_i^B(\overline{a}) \to \exists b \in A\ \varphi_j^B(\overline{a})$ and, since $A \subseteq B$, we also have the opposite, i.e., $\exists b \in A\ \varphi_j^B(\overline{a}) \to \exists b \in B\ \varphi_j^B(\overline{a}) \to \varphi_i^B(\overline{a})$. Then, by the induction hypothesis, $A \preceq_{\varphi_j} B$ and, therefore

$$\varphi_i^B(\overline{a}) \leftrightarrow \exists b \in A\ \varphi_j^B(\overline{a}) \leftrightarrow \exists b \in A\ \varphi_j^A(\overline{a}) \leftrightarrow \varphi_i^A(\overline{a})$$

and then $A \preceq_{\varphi_i} B$, as desired. ∎

We are now ready to state and prove the Reflection Theorem.

Theorem 3.3

(Reflection Theorem) Let $\varphi_1, \varphi_2, \ldots, \varphi_n \in \mathcal{L}$ and $B \neq \emptyset$ a class. If $A(\xi)$ satisfies

- $A(\xi) \subseteq A(\eta)$, whenever $\xi < \eta$;
- $A(\eta) = \bigcup_{\xi<\eta} A(\xi)$, for η a limit ordinal;
- $B = \bigcup_{\xi \in ON} A(\xi)$.

Then for all ξ, there is some limit $\eta > \xi$ such that $A(\eta) \neq \emptyset$ and $A(\eta) \preceq_{\varphi_i} B$, for $i = 1, \ldots, n$.

Clearly a lot of hierarchies and classes we saw satisfies this definition. In particular, assuming The Axiom of Regularity, $B = V$ and $A(\xi) = R(\xi)$ are natural candidates that we will use a lot.

Proof. The proof will follow similar steps as the proof of the Löwenheim-Skolem Theorem. Since we will use the previous lemma, we can assume WLOG that our list is subformula-closed. If not, replace each occurence of $\forall$ by $\neg\exists\neg$ and add each subformula of φ_i to the list.

As in Löwenheim-Skolem, for each $\varphi_i(\overline{v}) = \exists w\ \varphi_j(\overline{v}, w)$ define $F_i : B^n \to ON$ as follows: if $\varphi_i^B(\overline{a})$, let $F_i(\overline{a})$ be the least ζ such that $\exists b \in A(\zeta)\ \varphi_j^B(\overline{a}, b)$, otherwise let $F_i(\overline{a}) = 0$. These F_i are just as the skolem functions f_φ, but defined without AC, and saving the ordinal rank of a witness instead of the witness itself.

Now, we will make an analogous of the S_n's to store, instead of the witnesses for all parameters in S_{n-1}, the ordinal rank of all these witnesses, i.e., let $S : ON \to ON$ be given by

$$S(\alpha) = \sup\{F_i(\overline{a}) : i < n \text{ and } \overline{a} \in A(\alpha)\}$$

However, since we want our ordinal level η to be a limit ordinal such that $A(\eta) \neq \emptyset$, let's define the following increasing sequence: Given ξ, let η_0 be the least ordinal $\alpha > \xi$ such that $A(\alpha) \neq \emptyset$, and let $\eta_{n+1} = \max(\eta_n + 1, S(\eta_n))$. Then $\eta = \sup\{\eta_k : k \in \omega\}$ is a limit ordinal such that $A(\eta) \neq \emptyset$.

Now, given $\varphi_i(\overline{v}) = \exists b\ \varphi_j(\overline{v}, b)$ and $\overline{a} \in A(\eta)$, we have that $\overline{a} \in A(\eta_n)$ for some n. If $\varphi_i^B(\overline{a})$, then by definition there is some $b \in A(F_i(\overline{a}))$ such that $\varphi_j^B(\overline{a}, b)$, but $F_i(\overline{a}) \leq S(\eta_n) \leq \eta_{n+1} < \eta$ and, therefore, $\exists b \in A(\eta)\ \varphi_j^B(\overline{a}, b)$ so, by the previous Lemma, $A(\eta) \preceq_{\varphi_i} B$, for $i = 1, \ldots, n$. ∎

Observation: After the previous two results, you could be asking yourself where do we used that the set of formulas should be finite. If you keep reading, you will convince yourself that, if we could generalize these results to infinitely many formulas, we would have a lot of troubles.
The reason why a finite amount of formulas is the best we can do is because of Tarski's Theorems on the Definability and Undefinability of Truth. The former says roughly, in this context, that we can define using recursion, for each formula φ and set A, what does it means to $A \vDash \varphi$. In particular, this allows us to define what does it means to A satisfies Λ, where Λ is any set of formulas.
The latter, however, says roughly that we can't do the same for some classes. In fact, if some class V is such that $(V, \in) \vDash \text{ZFC}$, then being able to define $\vDash$ inside ZFC would allow it to prove its own consistency, which is impossible. In particular, we can't give a meaningful definition for $A \preceq V$, but we can for $A \preceq_\varphi V$, where φ is some formula, as we did in the previous results.
This has the same spirit as: if ZFC is consistent, then for all $n \in \omega$, ZFC proves that n is not a Gödel Number for a proof that $0 = 1$, why this doesn't implies that ZFC proves its own consistency? Because even if $\forall n \in \omega$, $\text{ZFC} \vdash \neg\text{Prov}(n, \ulcorner 0 = 1 \urcorner)$, we can't necessarily conclude that $\text{ZFC} \vdash \forall n \in \omega(\neg\text{Prov}(n, \ulcorner 0 = 1 \urcorner)) \equiv \text{Con(ZFC)}$. The toy model for this phenomenon is called ω-inconsistency, and it also explains a lot of interesting phenomena, like the fact that, if ZFC is consistent, then $\text{ZFC} + \neg\,\text{Con(ZFC)}$ is also consistent, despite the fact that it proves its own inconsistency.

Let's see one of the most useful application of this theorem

Corollary 3.1

Let $\Lambda \subseteq \text{ZF}$ be finite, then we can prove in ZFC that there is a ctm of $\text{ZC} \cup \Lambda$.

Proof. Assuming Regularity, let $B = V$ and $A(\xi) = R(\xi)$. If $\{\varphi_1, \varphi_2, \dots, \varphi_n\} = \Lambda$, apply the Reflection Theorem to get a limit $\eta > \omega$ such that $R(\eta) \preceq_{\varphi_i} V$, then $R(\eta) \vDash \text{ZC} \cup \Lambda$. Now, in order to get a ctm, just apply Mostowski Collapsing Lemma to $R(\eta)$. ■

This basically says that, even if ZFC cannot find a model for itself, it can find a model for every finite fragment of it. In some sense $R(\gamma)$ for suitable γ can approximate V satisfying ZC and every finite amount of instances of The Replacement Axiom.

Another interesting corollary is that ZF is not finitely axiomatizable, more generally:

Corollary 3.2

If $\Gamma \supseteq \text{ZF}$ is consistent, then Γ is not finitely axiomatizable.

Proof. Let $\Lambda \subseteq \Gamma$ be finite and define $\psi := \exists\eta(R(\eta) \vDash \Lambda)$, we may assume, expanding Λ if necessary, that Λ proves basic facts about $R(\eta)$ and $\vDash$. Then $\Gamma \vDash \psi$ by Reflection, but if $\Lambda \vdash \psi$, then Λ is inconsistent. This is clear by Gödel's Second Incompleteness Theorem, but we could've also defined η_0 as the least ordinal such that $R(\eta_0) \vDash \Lambda$, then $R(\eta_0) \vDash \Lambda + \nexists\eta(R(\eta) \vDash \Lambda)$, otherwise there would be some witness $\eta \in R(\eta_0)$, which is impossible since η_0 is the smallest one. ■

Since Λ is not necessarily a subset of ZF we can't explicitly use Reflection

We will now prove a set version of Reflection Theorem, obtained by replacing ON with a regular uncountable cardinal. Since we are dealing with sets, we can use infinitely many formulas, instead of a finite amount. This will be essential in the proof that $V = L$ implies GHC.

Theorem 3.4

Assuming AC, let $\kappa > \omega$ be a regular cardinal and $A(\xi)$ be such that, for $\xi \leq \kappa$:

1. $\xi < \eta \to A(\xi) \subseteq A(\eta)$;
2. $A(\eta) = \bigcup_{\xi<\eta} A(\xi)$ for limit $\eta \leq \kappa$;
3. $|A(\xi)| < \kappa$ for all $\xi < \kappa$, and $|A(\kappa)| = \kappa$.

Then for all ξ, there is some limit ordinal κ, with $\xi < \eta < \kappa$, such that $A(\eta) \neq \emptyset$ and $A(\eta) \preceq A(\kappa)$.

Proof. Pendent ∎

3.5 Constructible Sets

This subsection will be devoted to define one of the most important inner models of ZFC. In fact, it will be the minimal model of ZFC in some specific sense.

Definition 3.2

Let $P \subseteq A$, then $\mathcal{D}(A, P)$ is the set of all subsets of A that are definable over $(A, \in)$ with parameters in P. Then $\mathcal{D}^+(A) = \mathcal{D}(A, A)$ and $\mathcal{D}^-(A) = \mathcal{D}(A, \emptyset)$.

We will now give a sketch of the following important Lemma

Lemma 3.10

The function $\mathcal{D}(A, P)$ is absolute over transitive models $M \vDash \mathrm{ZF} - \mathcal{P}$.

Proof. Almost all the hard work use Theorem 3.2. First of all, the notion of φ being a formula is absolute, since we can define it recursively using absolute functions (in fact, this verification is the lame work). And the same can be done to "$\mathfrak{A} \vDash \varphi$", which is really boring, since this definition also involves a lot of intricate intermediate definitions we need to show to be absolute. But once we did that, we can then guarantee that $\mathcal{D}(A, P)$ is absolute, since it depends solely on $\vDash$. ∎

We can now define the Constructible Universe as a limit of a cumulative hierarchy

Definition 3.3

Define $L(\delta)$ by recursion on δ as follows:

- $L(0) = \emptyset$;
- $L(\beta + 1) = \mathcal{D}^+(L(\beta))$;
- $L(\gamma) = \bigcup_{\alpha<\gamma} L(\alpha)$ for γ a limit ordinal.

Then $L = \bigcup_{\alpha \in ON} L(\alpha)$.

3.5.1 $L \vDash$ ZF

This subsubsection will be devoted to develop the machinery needed to use Lemma 3.2 and conclude that L is in fact a model of ZF.

Let's start with the following basic properties of $L(\alpha)$

Lemma 3.11

For all ordinals α and β:

1. $L(\alpha) \subseteq R(\alpha)$;
2. $L(\alpha)$ is transitive;
3. $\alpha \leq \beta \to L(\alpha) \subseteq L(\beta)$;
4. $L(\alpha) \cap ON = \alpha$.

Proof. 1. Is pretty easy by induction on α, since $\mathcal{D}^+(A) \subseteq \mathcal{P}(A)$;

2. For $\alpha = 0$ or a limit ordinal it's clear from the definition. Then assume that $L(\alpha)$ is transitive then, given $A \in L(\alpha)$, we have that $A = \{x \in L(\alpha) : x \in A\} \in \mathcal{D}^+(L(\alpha)) = L(\alpha + 1)$. Now, given $x \in L(\alpha + 1)$, since $L(\alpha + 1) \subseteq \mathcal{P}(L(\alpha))$, then $x \subseteq L(\alpha) \subseteq L(\alpha + 1)$;

3. Fixing α, we can prove by induction on β. The cases $\beta = \alpha$ or a limit ordinal are trivial. The successor case follows by $L(\beta) \subseteq L(\beta + 1)$, that we proved for (2);

4. Also by induction, the base and limit cases are very straightforward. Assume then, as the induction hypothesis for the successor case, that $L(\beta) \cap ON = \beta$, then

$$\beta = L(\beta) \cap ON \subseteq L(\beta + 1) \cap ON \subseteq R(\beta + 1) \cap ON = \beta + 1$$

So it's sufficient to show that $\beta \in L(\beta + 1) \cap ON$. But, if $\varphi(x)$ is the Δ_0 formula equivalent to "x is an ordinal", then we have by absoluteness that

$$\beta = \{a \in L(\beta) : L(\beta) \vDash \varphi(a)\} \in \mathcal{D}^-(L(\beta)) \subseteq L(\beta + 1)$$

as desired. ■

As for the rank function in $R(\alpha)$ we can define the analogous L-rank function ρ for $L(\alpha)$.

Definition 3.4

For $x \in L$, the L-rank of x, $\rho(x)$, is the least α such that $x \in L(\alpha+1)$.

Then clearly we have that $L(\alpha) = \{x \in L : \rho(x) < \alpha\}$. Furthermore:

Lemma 3.12

For each α, we have $L(\alpha) \in L$ and $\rho(L(\alpha)) = \rho(\alpha) = \alpha$.

Proof. Clearly $L(\alpha) = \{x \in L(\alpha) : x = x\} \in D^+(L(\alpha)) = L(\alpha+1) \subseteq L$, and, since $L(\alpha) \notin L(\alpha)$, then $\rho(L(\alpha)) = \alpha$. By the previous lemma, we have that $\alpha \in L(\alpha+1) \setminus L(\alpha)$, then $\rho(\alpha) = \alpha$. ■

Another clear property is that every finite subset $\{a_1, a_2, \dots, a_n\}$ of $L(\alpha)$ is in $L(\alpha+1)$, we just need to consider $\varphi(y) = y = a_1 \vee y = a_2 \vee \cdots \vee y = a_n$. Then we can see when $R(\alpha)$ and $L(\alpha)$ start to diverge

Lemma 3.13

$R(\alpha) = L(\alpha)$ for all $\alpha \leq \omega$ and $L(\omega+1) \subset R(\omega+1)$.

Proof. By the previous observation $L(n) = R(n)$ for all $n \in \omega$. This then implies, by definition, that $L(\omega) = R(\omega) = HF$. However, $R(\omega+1) = \mathcal{P}(HF)$ is uncountable, while $L(\omega+1) = \mathcal{D}^+(HF)$ is countable, because there are only countably many formulas and HF is itself countable. ■

Observation: The previous lemma implies, for example, that $R(\omega+1) \setminus L(\omega+1)$ is uncountable. It's, however, pretty hard to give a "simple" example of an element in this set, the best we can do is that $\mathrm{Th}(HF) \in R(\omega+1) \setminus L(\omega+1)$, and $\rho(\mathrm{Th}(HF)) = \omega+1$, while $\mathrm{rank}(\mathrm{Th}(HF)) = \omega$.
Since formulas are encoded by sets in HF, clearly $\mathrm{Th}(HF) \in R(\omega+1)$. The fact that $HF \notin L(\omega+1) = \mathcal{D}^+(L(\omega))$ is a corollary of Tarski's Theorem on the undefinability of truth since, otherwise, truthiness on HF would be possibly defined within it. To show that $\mathrm{Th}(HF) \in L(\omega+2)$, however, is a bit tricky, we need to implement some new encoding inside $L(\omega+1)$, which we won't.

In order to calculate the size of $L(\alpha)$ we will need the following Lemma

Lemma 3.14

$|\mathcal{D}^+(A)| = |A|$ for all infinite A.

Proof. $|\mathcal{D}^+(A)| \leq |A|$, since $\{a\} \in \mathcal{D}^+(A)$ for all $a \in A$. Also, since there are $\aleph_0$ formulas and $|[A]^{<\omega}| = |A|$ parameters, then $|\mathcal{D}^+(A)| \geq \aleph_0 \cdot |A| = |A|$. ■

Theorem 3.5

Assuming AC, we have $|L(\alpha)| = |\alpha|$ for all $\alpha \geq \omega$.

Proof. Let's prove by induction. For $\alpha = \omega$, $L(\omega) = HF$ is clearly countable. Now, if $|L(\alpha)| = |\alpha|$, then, by the previous lemma $|L(\alpha+1)| = |\mathcal{D}^+(L(\alpha))| = |L(\alpha)| = |\alpha|$. For $\alpha = \gamma$ limit, since $\gamma \subseteq L(\gamma)$, then $|\gamma| \le |L(\gamma)|$. Also, if, for every $\beta < \gamma$, we have$|L(\beta)| = |\beta| \le |\gamma|$, then

$$|L(\gamma)| = \left|\bigcup_{\beta<\gamma} L(\beta)\right| \le \sum_{\beta<\gamma} |L(\beta)| \le |\gamma| \tag{2}$$

and then $|L(\gamma)| = |\gamma|$, as desired. AC is used in the first inequality of (2). ■

In fact, the use of AC in the previous proof is unnecessary, since we could just use absoluteness and redo the proof inside L, where, as we will see, satisfies AC.

In comparison, since $|R(\alpha)| = \beth_\alpha$ for great $\alpha \ge \omega^2$, while $|L(\alpha)| = |\alpha|$, it seems that $R(\alpha)$ is way bigger and grows way faster than $L(\alpha)$. It's then reasonable to expect that L should be way smaller than V, but, in fact, $V = L$ is consistent with ZF.

Theorem 3.6

L is a model for ZF.

Here is where we use almost all theorems we proved through this section.

Proof. By Lemma 3.2, it's enough to prove that $L \vDash \mathcal{C}$ and that for every $x \subseteq L$, there is a $y \in L$ such that $x \subseteq y$. For the latter, let $\alpha = \sup\{\rho(z) + 1 : z \in x\}$ and $y = L(\alpha)$.

For the former, let $\varphi(x, z, \overline{v})$ be a formula without y free, then $\mathcal{C}^L$ is equivalent to

$$\forall z, \overline{v} \in L\, \exists y \in L\, \forall x \in L\, [x \in y \leftrightarrow x \in z \wedge \varphi^L(x, z, \overline{v})]$$

So, given $z, \overline{v} \in L$, let $y = \{x \in z : \varphi^L(x, z, \overline{v})\}$, we just need to show that $y \in L$. Fix ξ such that $z, \overline{v} \in L(\xi)$, since $L(\alpha)$ is a cumulative hierarchy, applying the Reflection Theorem to φ, there is a limit $\eta > \xi$ such that $L(\eta) \preceq_\varphi L$, and then

$$y = \left\{x \in L(\eta) : (x \in z \wedge \varphi(x, z, \overline{v}))^{L(\eta)}\right\} \in \mathcal{D}^+(L(\eta)) = L(\eta+1) \subseteq L$$

■

3.6 The Axiom of Constructibility $V = L$

Let's consider now the Axiom of Constructibility $V = L$, which is actually just a shorthand for $\forall x\, \exists \alpha \in ON\, [x \in L(\alpha)]$. In order to see wether it's true in M, we need to compute $L(\alpha)^M$, fortunately, it's absolute over almost all models we will consider, as the next lemma shows

Lemma 3.15

The function $\alpha \mapsto L(\alpha)$ is absolute for transitive $M \vDash \mathrm{ZF} - \mathcal{P}$.

Proof. Since we are working over $\mathrm{ZF} - \mathcal{P}$ the construction of $L(\alpha)$ is guaranteed to exist. Now, by Lemma 3.2 ordinals are absolute over M and, by Lemma 3.5 $\mathcal{D}^+(A)$ is also absolute. By induction, clearly $L(0)^M = L(0)$, if $L(\alpha)^M = L(\alpha)$, then $L(\alpha+1)^M = \mathcal{D}^+(L(\alpha))^M = \mathcal{D}^+(L(\alpha)) = L(\alpha+1)$ and, if $L(\xi)^M = L(\xi)$ for all $\xi < \eta$, where η is a limit ordinal, then

$$L(\eta)^M = \left(\bigcup_{\xi<\eta} L(\xi)\right)^M = \bigcup_{\xi<\eta} L(\xi) = L(\eta)$$

since, as we saw in Lemma 3.2, $\bigcup$ is also absolute. ■

Corollary 3.3

$(V = L)^L$

Proof. We need to verify $(\forall x\, \exists \alpha \in ON\, [x \in L(\alpha)])^L$ but, since $ON \subseteq L$ is absolute, then

$$(\forall x\, \exists \alpha \in ON\, [x \in L(\alpha)])^L \equiv \forall x \in L\, \exists \alpha \in ON\, [x \in L(\alpha)]$$

which is clearly true by the definition of L. ■

Although this should seem trivial, it's not. Note how in the previous lemma we used Lemma 3.5 which we don't even proved, just showed a sketch. In fact, there are a lot of notions we need to prove to be absolute before we could prove that $(V = L)^L$.

Definition 3.5

For M a transitive set $o(M) = M \cap ON$ is the first ordinal not in M.

The absoluteness of $L(\alpha)$ lets us identify the transitive set models of $V = L$

Lemma 3.16

Let $M \vDash \mathrm{ZF} - \mathcal{P}$ be a transitive set. Then $M \vDash V = L$ iff $M = L(o(M))$.

Proof. Let $\gamma = o(M)$. By absoluteness, $L(\alpha) \in M$ for all $\alpha < \gamma$ and, since M is transitive, $L(\gamma) \subseteq M$. Also $(V = L)^M \equiv \forall x \in M\, \exists \alpha \in \gamma\, [x \in L(\alpha)]$, i.e., $M \subseteq L(\gamma)$, then $M = L(\gamma)$. ■

Now, in order to verify that $(\mathrm{AC}+\mathrm{GHC})^L$, we will devote the following subsubsections to the proof that $V = L$ implies both AC (in fact, it implies Global Choice) and GHC.

3.6.1 $V = L$ Implies Global Choice

Since AC is equivalent to every set can be well ordered, and $V = L$, it's sufficient to show each $L(\alpha)$ can be well ordered. We will first show how to well order $\mathcal{D}^+(A)$, whenever A can also be.

Intuitively, if A can be well ordered, then also $A^{<\omega}$, and clearly the countable set of formulas also can. Then, given $S_i = \{a \in A : A \vDash \varphi_i(\bar{b}_i, a)\} \in A$, for $i = 1, 2$, we first compare φ_1 with φ_2 and, if they are equal, then $\bar{b}_1$ with $\bar{b}_2$. I will not give the full proof, just a sketch.

I will first assume that the countable set of all $\in$-formulas can be well-ordered, which is actually not a hard thing to do, it's just really boring. Let's call this well ordering $<_F$ for Formulas.

Lemma 3.17

If (A, R) is a well order, then there is a well order $<_{(A,R)}$ for $\mathcal{D}^+(A)$.

Sketch: First of all, if A is well ordered by R, then A^n can also be well ordered, for each $n \in \omega$, just order it using the R-lexicographic order $<_{A^n}$. Now, in order to formalize our previous idea, given S_i we know that both φ_i and $\bar{b}_i$ aren't necessarily unique, then, for each $S \in \mathcal{D}^+(A)$, let $(\varphi, \bar{b})$ be the least pair in the lexicographic order induced by $<_F$ and $<_{A^n}$ such that $S = \{a \in A : A \vDash \varphi(\bar{b}, a)\}$. Then, given $S_1, S_2 \in \mathcal{D}^+(A)$ as before, with $(\varphi_1, \bar{b}_1)_{S_1}$ and $(\varphi_2, \bar{b}_2)_{S_2}$, let

$$S_1 <_{(A,R)} S_2 \iff (\varphi_1 <_F \varphi_2) \text{ or } (\varphi_1 = \varphi_2 \wedge \bar{b}_1 <_{A^n} \bar{b}_2)$$

We can then well order each $L(\alpha)$ (and therefore L) as following

Definition 3.6

By recursion on α, let $<_\alpha \subseteq L(\alpha) \times L(\alpha)$ be defined as: for $x, y \in L(\alpha)$, let $x <_\alpha y$ iff $\rho(x) < \rho(y)$ or $\rho(x) = \rho(y) = \beta < \alpha$ and $x <_{(L(\beta), <_\beta)} y$. Then, let $<_L$ be defined as:

$$x <_L y \iff \rho(x) < \rho(y) \text{ or } (\rho(x) = \alpha = \rho(y) \wedge x <_\alpha y)$$

Theorem 3.7

$<_L$ well orders L and, furthermore, each $L(\alpha)$ is an initial segment, with

$$<_\alpha = <_L \cap (L(\alpha) \times L(\alpha))$$

Now, assuming $V = L$, we have that $<_L$ well orders V and, therefore, AC holds.

We actually have something stronger, since $<_L$ well orders everything, then $V = L$ implies Global Choice.

3.6.2 $V = L$ Implies GHC

The following lemma will be surprisingly useful in the proof that $V = L$ implies GHC.

Lemma 3.18

If κ is any regular uncountable cardinal, then $L(\kappa) \vDash \text{ZF} - \mathcal{P} + V = L$.

Proof. Pendent ∎

The similarity of the proofs for $L(\kappa)$ and $H(\kappa)$ for $\kappa \geq \omega$ is no coincidence, since:

Theorem 3.8

$V = L$ implies that $L(\kappa) = H(\kappa)$ for all cardinals $\kappa \geq \aleph_0$. Hence $V = L \rightarrow \text{GHC}$.

Proof. First of all, since we showed $L(\omega) = HF = H(\omega)$, we didn't need to consider the base case. Also, If $x \in L(\kappa)$ for some $\kappa > \aleph_0$, fix a cardinal $\omega \leq \lambda < \kappa$ such that $x \in L(\lambda)$, then $\text{trcl}(x) \subseteq L(\lambda)$ and, therefore, $|\text{trcl}(x)| \leq |L(\lambda)| = \lambda < \kappa$, so $x \in H(\kappa)$ and $L(\kappa) \subseteq H(\kappa)$ for all $\kappa > \aleph_0$.

Then, it's sufficient to show that $L(\kappa) \subseteq H(\kappa)$ for all $\kappa > \aleph_0$. In fact, note that we actually just need to prove it for successor cardinals, since, when κ is an uncountable limit cardinal, we have

$$L(\kappa) = \bigcup_{\lambda<\kappa} L(\lambda^+) = \bigcup_{\lambda<\kappa} H(\lambda^+) = H(\kappa)$$

Let λ be an infinite cardinal, fix some $b \in H(\lambda^+)$ and let $T = \{b\} \cup \text{trcl}(b)$. Then $b \in T$ and $|T| \leq \lambda$. Pendent

Then, finally, if $H(\kappa) = L(\kappa)$, then, by Theorems 1.2.2 and 3.5.1 we have that

$$2^\kappa \leq 2^{<\kappa^+} = |H(\kappa^+)| = |L(\kappa^+)| = \kappa^+ \leq 2^\kappa$$

for all cardinals $\kappa \geq \aleph_0$, which is exactly what GHC states. ■

CHAPTER 4

Independence of AC Without Forcing

4.1 Ordinal Definable Sets

Pendent

4.2 Rieger-Bernays Permutation Model

In order to give a different proof for the consistency of $\mathrm{ZF}^- + \neg\mathcal{R}$, we will study permutations of the whole universe V and define a new $\in$-relation with this permutation.

Definition 4.1

Let $F : V \to V$ be a function. Define $E = E_F \subseteq V \times V$ by xEy iff $x \in F(y)$.

As usual, F is just a formula $\psi(x, y)$ where $\forall x\, \exists! y\, \psi(x, y)$ and xEy abbreviates $\exists y\, [\psi(x, y) \wedge x \in y]$.

We can now prove a really interesting theorem which gives us a lot of different models for ZFC

Theorem 4.1

Let $F : V \to V$ be a permutation of V. Then $\varphi^{(V,E)}$ holds for each $\varphi \in \mathrm{ZF}^-$.

Proof.

- First, for Extensionality we have that

$$\mathcal{E}^{(V,E)} \equiv \forall x, y\, (x = y \leftrightarrow \forall z(zEx \leftrightarrow zEy))$$

substituting the definition of E, we have that

$$\mathcal{E}^{(V,E)} \equiv \forall x, y(x = y \leftrightarrow \forall z(z \in F(x) \leftrightarrow z \in F(y)) \equiv \forall x, y(x = y \leftrightarrow F(x) = F(y))$$

which says that F is a bijection, that is guaranteed by the fact that it's a permutation;

- For the Axiom of Union, we hve that

$$\begin{aligned}\mathcal{U}^{(V,E)} &\equiv \forall A\, \exists U\, \forall y(yEU \leftrightarrow \exists z(yEz \wedge zEA))\\ &\equiv \forall A\, \exists U\, \forall y(y \in F(U) \leftrightarrow \exists z(y \in F(z) \wedge z \in F(A)))\end{aligned}$$

  Which basically says that $F(U) = \bigcup_{z \in F(A)} F(z)$, then, seting

$$U = F^{-1}\left(\bigcup_{z \in F(A)} F(z)\right)$$

  is sufficient to prove it.

- For the Axiom of Powerset, we need to interpret $(x \subseteq y)^{(V,E)}$, which is $\forall w(wEx \to wEy)$, i.e., $\forall w(w \in F(x) \to w \in F(y))$, in other words, $F(x) \subseteq F(y)$. Therefore

$$\begin{aligned}\mathcal{P}^{(V,E)} &\equiv \forall x\, \exists y\, \forall z\Big(zEy \leftrightarrow (z \subseteq x)^{(V,E)}\Big)\\ &\equiv \forall x\, \exists y\, \forall z(z \in F(y) \leftrightarrow F(z) \subseteq F(x))\end{aligned}$$

  then, we can let y be such that $F(y) = \{z : F(z) \subseteq F(x)\}$. In order to prove it's a set, just note that $F(y) = F^{-1}(\mathcal{P}(F(x)))$, which is a set. Then set $y = F^{-1}(F^{-1}(\mathcal{P}(F(x))))$.

- For the Axiom of Infinity, we need to relativize $\emptyset$ and the S function in (V, E). First, note that $y = \emptyset$ iff $\forall z(z \in y \leftrightarrow z \neq z)$, i.e., relativized to (V, E), $\forall z(z \in F(y) \leftrightarrow z \neq z)$, then $(y = \emptyset)^{(V,E)}$ iff $F(y) = \emptyset$, so $y = F^{-1}(\emptyset)$. Analogously, $y = S(x)$ iff $\forall z(z \in y \leftrightarrow (z \in x \vee z = x))$, then, in (V, E), we have that $(y = S(x))^{(V,E)}$ iff $\forall z(z \in F(y) \leftrightarrow (z \in F(x) \vee z = x))$, equivalently, $F(y) = F(x) \cup \{x\}$. Therefore

$$\begin{aligned}\mathcal{I}^{(V,E)} &\equiv \exists x(\emptyset^{(V,E)}Ex \wedge \forall y(yEx \to S(y)^{(V,E)}Ex))\\ &\equiv \exists x(F^{-1}(\emptyset) \in F(x) \wedge \forall y(y \in F(x) \to F(y) \cup \{y\} \in F(x)))\end{aligned}$$

  we can then define $A = \bigcup_{n \in \omega} A_n$ by recursion, where $A_0 = \{F^{-1}(\emptyset)\}$ and, for all $n \geq 0$

$$A_{n+1} = A_n \cup \{F(x) \cup \{x\} : x \in A_n\}$$

  Then let $F(x) = A$, i.e., $x = F^{-1}(A)$.

- Finally, for the Axiom Schema of Replacement, clearly $\mathcal{F}^{(V,E)}$ will state that, if $\varphi^{(V,E)}(x, y, \overline{v}, A)$ is "a function" in $F(A)$, then there is a B such that $F(B) = \mathrm{Im}(\varphi^{(V,E)}(x, y, \overline{v}, A))$. In fact, since $F(A)$ is a set, we can just apply replacement and then let $B = F^{-1}(\mathrm{Im}(\varphi^{(V,E)}(x, y, \overline{v}, A)))$.

■

Note how powerful the previous theorem is. It's basically saying that, assuming consistency of ZF^-, for every model V of it we can make a new one by just choosing a permutation F in it. Let's see how such basic permutations gives us impressive results.

Corollary 4.1

If ZF is consistent (or even ZF^-), then $\mathrm{ZF}^- + \neg\mathcal{R}$ is also consistent.

Proof. Let $(V, \in) \vDash \mathrm{ZF}$, and consider F as the transposition of 0 and 1. Then, for any set x, we have that $xE0$ iff $x \in F(0) = 1$, i.e., $x = 0$, then $y = 0$ proves that $(V, E) \vDash \exists y(y = \{y\})$. ■

Observation: Such permutation F is called the Rieger-Bernays Permutation, and the model (V, E_F) is called the Rieger-Bernays Permutation model. As we saw, it introduces an element $x = \{x\}$, such elements are called Quine Atoms, and it's not hard to see that $x = 0$ is the only one in this model.

4.3 Consistence of ¬AC

Pendent

CHAPTER 5

Martin's Axiom

In the 60's, Martin, Solovay, Tennenbaum, and others extended Cohen's method of forcing and proved many statements to be independent of ZFC + $\neg$CH. But then Martin and a few other people realized that one could encapsulate the results of some of these proofs into one axiom, now called Martin's Axiom or MA, which would have been understandable in the 20's. It's consistent with it that $\mathfrak{c}$ is arbitrarily large, and in fact it's a consequence of CH. MA is usually covered before forcing, since it introduces many of the combinatorial complexities found there, as we will see.

5.1 Forcing Posets

Definition 5.1

A forcing poset is a triple $(\mathbb{P}, \leq, \mathbb{1})$, where $\leq$ is a pre order on $\mathbb{P}$ and $\mathbb{1}$ is a maximum element in $\mathbb{P}$.

Elements of $\mathbb{P}$ are called forcing conditions, and $p \leq q$ is read "p extends q".

Definition 5.2

Let $\mathbb{P}$ be a forcing poset, then $p, q \in \mathbb{P}$ are compatible ($p \not\perp q$) iff they have a common extension, and incompatible ($p \perp q$) otherwise. $A \subseteq \mathbb{P}$ is an antichain iff its elements are pairwise incompatible. $\mathbb{P}$ has the countable chain condition (ccc) iff every antichain in $\mathbb{P}$ is countable.

The following example gives us some intuition on how the elements of $\mathbb{P}$ are approximations of something we are trying to construct.

Definition 5.3

For any I, J, let $\mathrm{Fn}(I, J)$ be te set of all finite partial functions from I to J. Make it into a forcing poset by letting $\leq$ be $\supseteq$ and $\mathbb{1} = \emptyset$.

So $p \not\perp q$ iff p, q are compatible as function, i.e., they have a common extension or, equivalently, $p \cup q \in \mathrm{Fn}(I, J)$. Informally, each $p \in \mathrm{Fn}(I, J)$ is a finite approximation of some function $f : I \to J$, then it informally expresses a condition that our f needs to satisfy, namely $f \supseteq p$.

In order to define MA, we need the following concepts

Definition 5.4

Let $\mathbb{P}$ be a forcing poset. Then $D \subseteq \mathbb{P}$ is dense in $\mathbb{P}$ iff $\forall p \in \mathbb{P}\, \exists q \in D\; q \leq p$.

Let's see examples of dense sets in $\mathbb{P}$ using the previous forcing poset.

Example: If I is infinite and $J \neq \emptyset$, then

$$D_i = \{q \in \mathrm{Fn}(I, J) : i \in \mathrm{dom}(q)\} \text{ and } I_j = \{q \in \mathrm{Fn}(I, J) : j \in \mathrm{Im}(q)\}$$

are dense whenever $i \in I$ and $j \in J$.
As another example, if $|J| \geq 2$, then

$$E_h = \{q \in \mathrm{Fn}(I, J) : q \nsubseteq h\}$$

is dense, whenever $h \in J^I$.

Which is clear, since, for the first, any finite partial function can be extended to one that contains i in its domain or j in its image; and, for the second, let $i \in I \setminus \mathrm{dom}(p)$ and $j \in J$, with $j \neq h(i)$, then $q = p \cup \{(i, j)\}$ extends p and is in E_h.

Definition 5.5

Let $\mathbb{P}$ be a forcing poset. Then $G \subseteq \mathbb{P}$ is a filter on $\mathbb{P}$ iff

1. $\mathbb{1} \in G$;
2. $\forall p, q \in G\, \exists r \in G\, [r \leq p \wedge r \leq q]$;
3. $\forall p, q \in \mathbb{P}\, [p \geq q \wedge q \in G \to p \in G]$

Given a family $\mathcal{D}$ of dense subsets of $\mathbb{P}$, a filter G is called $\mathcal{D}$-generic iff $G \cap D \neq \emptyset$ for every $D \in \mathcal{D}$.

In some sense, filters is a measure for "largeness". Condition 1 says $\mathbb{1}$ is large, condition 2. says that any common extension of larger elements is also large, and condition 3. says G is upward closed, i.e., given any large element, every other greater one is also large.

5.2 MA(κ) and Generic Filter Existence Lemma

We can now state Martin's Axiom. In fact, since it's based on forcing arguments, the reasons for why we should accept it or why these weird concepts like generic filters, dense and ccc forcing posets appears will be clarified in the next section where we explain the forcing technique.

Definition 5.6

$\mathrm{MA}_{\mathbb{P}}(\kappa)$ is the statement that whenever $\mathcal{D}$ is a family of dense subsets of $\mathbb{P}$, with $|\mathcal{D}| \leq \kappa$, there exists a $\mathcal{D}$-generic filter G on $\mathbb{P}$.

- $\mathrm{MA}(\kappa)$ is the statement that $\mathrm{MA}_{\mathbb{P}}(\kappa)$ holds for all ccc posets $\mathbb{P}$;
- MA is the statement that $\mathrm{MA}(\kappa)$ holds for every $\kappa < \mathfrak{c}$

From the definition, clearly $\mathrm{MA}_{\mathbb{P}}(\kappa) \to \mathrm{MA}_{\mathbb{P}}(\lambda)$ and $\mathrm{MA}(\kappa) \to \mathrm{MA}(\lambda)$, whenever $\lambda \leq \kappa$.

Now, motivated by the definition and our previous example, let's prove the following lemma.

Lemma 5.1

$\mathrm{Fn}(I, J)$ has the ccc iff $I = \emptyset$ or J is countable.

Proof. Pendent ∎

We can then illustrate how we can force some desired properties with the following lemma.

Lemma 5.2

if $\mathrm{MA}(\kappa)$, then $\kappa < \mathfrak{c}$.

Proof. Since $\mathrm{MA}(\kappa)$, in particular $\mathrm{MA}_{\mathbb{P}}(\kappa)$, for $\mathbb{P} = \mathrm{Fn}(I, J)$, where $I = \emptyset$ or J is countable. Now, if G is any filter on $\mathbb{P}$, given $p, q \in G$, since $p \not\perp q$ then they agree on $\mathrm{dom}(p) \cap \mathrm{dom}(q)$, so $f_G = \bigcup G$ is a function with $\mathrm{dom}(f_G) \subseteq I$ and $\mathrm{Im}(f_G) \subseteq J$.

If we force $f_G : I \to J$ to be surjection, but also force that $f \notin J^I$ we will have a contradiction. In order to force these conditions, we need to find some family of dense sets $\mathcal{D}$, and the fact that G is $\mathcal{D}$-dense will force f_G to have the desired properties.

First of all, to guarantee that $\mathbb{P}$ is pcc, let's take J countable as in the previous lemma. Now, to force the first condition, we know by the previous example that, if $J \neq \emptyset$, D_i and R_j are dense, then, if G intersects all these sets, clearly $f_G \in J^I$. For the second condition, if $|J| \geq 2$, then E_h is dense, which means that, if G intersects all E_h, then $f_G \neq h$ for all $h \in J^I$.

In fact, if $\mathrm{MA}(\kappa)$, for $\kappa \geq \mathfrak{c}$, then, for $|I| = \aleph_0$ and $|J| \geq 2$, we have

$$\mathcal{D} = \{D_i, R_j, E_h : i \in I, j \in J, h \in J^I\}$$

as a family of dense sets in $\mathbb{P}$. Since $|\mathcal{D}| \leq |J^I| \leq \aleph_0^{\aleph_0} = \mathfrak{c} \leq \kappa$ then, by $\mathrm{MA}(\kappa)$, we have that there is a $\mathcal{D}$-generic filter G and, then, $f_G = \bigcup G$ is such that $f \in J^I$, but $f \neq h$ for all $h \in J^I$. ∎

The previous proof shows us why we need $\mathbb{P}$ to be ccc:

Lemma 5.3

For all $\kappa \geq \aleph_1$, there is a non-ccc $\mathbb{P}$ such that $\mathrm{MA}_{\mathbb{P}}(\kappa)$ is false.

Proof. Let $\mathbb{P} = \mathrm{Fn}(I, J)$, where $|I| = \aleph_0$ and $|J| = \kappa$, then $\mathcal{D} = \{D_i, R_j : i \in I, j \in J\}$ has cardinality $\leq \kappa$ and, then, any $\mathcal{D}$-generic filter G would give us a surjective $\bigcup G = f_G : I \to J$. ∎

It doesn't work for $\kappa = \aleph_0$ since, in fact, we can prove MA($\aleph_0$) in ZFC using the following lemma, which will be of great importance in the next section when talking about forcing.

Lemma 5.4

(**Generic Filter Existence Lemma**) Let $\mathbb{P}$ be a ny forcing poset and $\mathcal{D}$ a countable family of dense subsets of $\mathbb{P}$. For any $p \in \mathbb{P}$ there is a $\mathcal{D}$-generic filter G on $\mathbb{P}$ such that $p \in G$.

Sketch: The idea is that filters can be generated by chains (subset where all elements are pairwise compatible). In fact, if C is any chain, then $G = \{q \in \mathbb{P} : \exists p \in C(p \leq q)\}$ is a filter, since $\mathbb{1} \in G$, for all $p, q \in G$, by definition there are $r, s \in C$ such that $r \leq p$ and $s \leq q$, in fact, since C is a chain, we can assume WLOG that $r \leq s$, then $r \leq p, q$. And, if $p, q \in \mathbb{P}$ with $p \geq q$ and $q \in G$, then $p \geq q \geq r$ for some $r \in C$ and, therefore, $p \in G$.

Proof. Motivated by the sketch, let $\mathcal{D} = \{D_n : n \in \omega\}$. Using AC and recursion on ω we will construct a chain $\{r_n\}_{n\in\omega}$ of elements in each D_n starting with p. Let then $r_0 = p$ and $r_{n+1} \leq r_n$ be such that $r_{n+1} \in D_n$, in fact such r_{n+1} does exists, since D_n is dense.

Then $C = \{r_n : n \in \omega\}$ is a chain and, as we saw, $G = \{q \in \mathbb{P} : \exists n(r_n \leq q)\}$ is a filter. Clearly $r_0 = p \in G$ and, since each $r_{n+1} \in G$, then $G \cap D_n \neq \emptyset$ and G is $\mathcal{D}$-generic. ■

In particular, CH would imply MA. Note that we can't prove an analogous lemma for $\mathcal{D}$ indexed by any larger ordinal α, since the definition of r_λ for λ a limit ordinal would naturally be such that $r_\lambda \leq r_\xi$ for all $\xi < \lambda$, but we couldn't use density to guarantee that such r_λ in fact exists.

5.3 Atomless and Centered Posets

Definition 5.7

$r \in \mathbb{P}$ is an atom iff there are no $p, q \leq r$ such that $p \perp q$. We say $\mathbb{P}$ is atomless iff there are no atoms in it.

Let's see why we care about atoms

Lemma 5.5

If $\mathbb{P}$ has an atom, then $G = \{q \in \mathbb{P} : q \not\perp r\}$ is a filter that meets all dense sets in $\mathbb{P}$.

Proof. If r is an atom, then, clearly $\mathbb{1} \in G$, since $r \leq \mathbb{1}, r$, i.e., $\mathbb{1} \not\perp r$, also, if $p, q \in G$, then there are $a \leq p, r$ and $b \leq q, r$; since r is an atom we have $a \not\perp b$, so there is some $c \leq a, b \leq p, q$ and, since $c \leq c, r$, then $c \not\perp r$, i.e., $c \in G$. Lastly, if $p \geq q$ and $q \in G$, then there is some $s \leq q, r$, so $s \leq p$ and $p \not\perp r$ too, i.e., $p \in G$. This guarantees that G is a filter but, furthermore, given $D \subseteq \mathbb{P}$ dense, there is some $p \in D$ such that $p \leq r$, then $p \not\perp r$, i.e., $p \in G \cap D$, and then G intersects every dense set. ■

Since the previous lemma just gave us a free witness for $\mathrm{MA}_{\mathbb{P}}(\kappa)$, we have the following corollary.

Corollary 5.1

If $\mathbb{P}$ has an atom, then $\mathrm{MA}_{\mathbb{P}}(\kappa)$ holds for all κ. If $\mathbb{P}$ is atomless, then $\neg\mathrm{MA}_{\mathbb{P}}\big(2^{|\mathbb{P}|}\big)$.

Proof. The first one is clear by the previous Lemma. For the second, let $\mathbb{P}$ be atomless and G be any filter. Then $D := \mathbb{P} \setminus G$ is dense, since whenever $r \in \mathbb{P}$, r is not an atom, so there are $p, q \leq r$ such that $p \perp q$, thus p, q are not both in G, so at least one of these must be in $\mathbb{P} \setminus G = D$, then G cannot meet D and, in particular, it cannot meet all dense sets of $\mathbb{P}$, whose amount is $\leq 2^{|\mathbb{P}|}$. ■

5.4 Applications

As we said before, the reason why we would care about concepts like density, generic filters, ccc, etc. will be clear when we motivate forcing. For now, we're trying to explain how posets would appear naturally.

The idea to approximate our generic object G by things p, q that *could be true* about it. So it ends up to be something like an implication $p \to q$ meaning that if p is true about G, then also is q. And this is exactly a poset, where $p \to q$ is interpreted as $p \geq q$.

It's worth noting that MA is just a single axiom (the most known one) in a large variety of others called Forcing Axioms. In the previous section, we just saw how to use it to approximate objects using one kind of poset / forcing notion, known as Cohen Forcing (or sometimes Easton Forcing).

5.4.1 Cardinal Characteristics

Cardinal Characteristics of the Continuum is a vast subarea of combinatorial set theory that analyzes combinatorial defined cardinals that are consistently between $\aleph_1$ and $\mathfrak{c}$, that's why the name.

One of the greatest achievements of Cardinal Characteristics is the so called Cichoń's Diagram:

Picture and explain it

Now, let's ramble a lit bit about cardinal characteristics κ. Since they are such that $\aleph_1 \leq \kappa \leq \mathfrak{c}$ and are consistently in between, it's expected that they have a lot to do with the Real numbers. In fact, the same happens with MA, that can be seen as a generalized Baire's Theorem.

As we saw in the diagram, a lot of cardinals there are measure-topological defined in terms of $\mathbb{R}$. However, some of them like $\mathfrak{b}$ and $\mathfrak{d}$, and a lot of others can be defined combinatorially and, if you look at all those definitions, they are usually defined in terms of properties that *almost happens*, meaning that happen for all, but a finite amount of terms. We can see this in concepts like dominating $\geq^*$, almost contained $\subseteq^*$ and almost disjoint relations.

Such *almost something* properties happen to be really fruitful when using specific forcing notions to approximate our objects, and this is what we will start to see now. When we get comfortable with this, we will step up to start approximate more complicated objects.

5.4.2 Hechler Forcing

Definition 5.8

Given $f, g \in \omega^\omega$, let $f \geq^* g$, whenever $f(n) \geq g(n)$ for all, but finitely many $n \in \omega$.
A family $\mathcal{F} \subseteq \omega^\omega$ is called unbounded if there is no single $g \in \omega^\omega$ such that, for all $f \in \mathcal{F}$, $g \geq^* f$.
The dominating number $\mathfrak{b}$ is the least cardinality of an unbounded family.

We will see that MA(κ) implies that $\mathfrak{b} > \kappa$ by defining some forcing notion and, using our G, create such dominating function g.

Theorem 5.1

Let $\kappa \geq \aleph_0$. Under MA(κ), we have $\mathfrak{b} > \kappa$. I.e., if $\mathcal{F} \subseteq \omega^\omega$, with $|\mathcal{F}| \leq \kappa$, then there is some $g \in \omega^\omega$ such that $g \geq^* f$, for all $f \in \mathcal{F}$, so $\mathcal{F}$ is not unbounded.

Sketch: The forcing notion we will sketch here is called Hechler forcing. Since our generic object will be a function $g \in \omega^\omega$, we can, as in the previous forcing notions, try to approximate it using finite partial functions $s \in \omega^{<\omega}$.
The problem is, now we have some family $\mathcal{F}$ of conditions to add. In order to approximate such conditions, we can, as some analogous attempt of approximate functions with its finite segments, approximate our conditions $\mathcal{F}$ as finite subsets $F \in [\mathcal{F}]^{<\omega}$.
Then our forcing notion $\mathbb{P}$ will have pairs (s, F) and, for a suitable $\mathcal{D}$, our $\mathcal{D}$-generic filter G will give rise to a function $g = \bigcup\{s : \exists F\,(s, F) \in G\} \in \omega^\omega$. To guarantee $g : \omega \to \omega$, we can use a similar set of dense sets D_n as in Easton's Forcing. To guarantee $g \geq^* f$ for every $f \in \mathcal{F}$, we want that G have at least one pair (s, F), where $f \in F$, so the most natural choice for the dense sets E_f are such pairs, where $f \in F$.
Now, it remains to define our order $\leq$. Naturally, we want in $(s, F) \leq (t, H)$ that, for example, $t \subseteq s$ and $H \subseteq F$, but, furthermore, it would be nice if s dominates every $h \in H$, so when we start to grow our chain $\cdots \leq (s_n, F_n) \leq \cdots \leq (s_2, F_1) \leq (s_1, F_1)$, each s_m dominates all previous $f \in F_m$.
Two things to note about this, first, this will not guarantee us that $(s, F) \leq (s, F)$, since we would need to have s dominating each $f \in F$ previously. And second, we don't even need all this complication. Given $f \in \mathcal{F}$, since $g \geq^* f$, there is some n_0 such that $g(n) \geq f(n)$, for all $n \geq n_0$. We could assume that our pair (s, F) that G founds, where $f \in F$, is such that $n_0 \in \mathrm{dom}(s)$, meaning the extensions we want for s is all such that just the next numbers added in the domain dominates the elements of F.

Proof. Based in our messy previous sketch, let's define

$$\mathbb{P} = \{(s, F) : s \in \omega^{<\omega} \text{ and } F \in [\mathcal{F}]^{<\omega}\}$$

and define our relation $\leq$ as following: Given $(s, F), (t, H) \in \mathbb{P}$

$$(s, F) \leq (t, H) \iff s \supseteq t,\ F \supseteq H, \text{ and, for all } h \in H,\ s(n) \geq h(n),\ \forall n \in \mathrm{dom}(s) \setminus \mathrm{dom}(t)$$

It's routine to verify that $(\mathbb{P}, \leq)$ is a poset. Let's now define, for each $n \in \omega$ and $f \in \mathcal{F}$:

$$D_n = \{(s, F) \in \mathbb{P} : n \in \mathrm{dom}(s)\} \quad \text{and} \quad E_f = \{(s, F) \in \mathbb{P} : f \in F\}$$

To see they are dense, let $(s, F) \in \mathbb{P}$. If $n \in \operatorname{dom}(s)$, then $(s, F) \in D_n$ already, otherwise, let $m = \max_{f \in F} f(n)$, then $D_n \ni (s \cup \{(n, m)\}, F) \leq (s, F)$. Analogously, $E_f \ni (s, F \cup \{f\}) \leq (s, F)$.

So, let $\mathcal{D} = \{D_n, E_f : n \in \omega, f \in [\mathcal{F}]^{<\omega}\}$, since $|\mathcal{D}| \leq \omega \cdot \kappa = \kappa$, it just remains to show it's ccc. Let's skip it by now and use $\mathrm{MA}(\kappa)$ to create our $\mathcal{D}$-generic filter G over $\mathbb{P}$.

Let then $g = \bigcup\{s : \exists F\, (s, F) \in G\}$. Given $n \in \omega$, there is some $(s, F) \in G \cap D_n$, so $n \in \operatorname{dom}(g)$, and $g : \omega \to \omega$, as desired. Now, given $f \in \mathcal{F}$, there is some $(s, F) \in G \cap E_f$, let $\operatorname{dom}(s) = n_0$, we claim that $g(n) \geq f(n)$, $\forall n > n_0$. In fact, let $n > n_0$ and $(t, H) \in G \cap D_n$, then, since G is a filter, there is some $(p, J) \leq (s, F), (t, H)$, so $p(n) \geq f(n)$, as desired.

The reason to g be well defined is that $(s, F) \not\perp (t, H)$ guarantees us s and t are compatible. ■

Now, as we said, the proof that $(\mathbb{P}, \leq)$ is ccc was skipped. In fact, based on the argument we needed to use to prove Easton's Forcing was ccc, we expect it to be as hard, using some hard combinatorial lemma as the Δ-System Lemma.

In fact, it turns out that it's very easy to show $(\mathbb{P}, \leq)$ is ccc, by just using a counting argument.

Before seeing it, let's define the more general scenario.

Definition 5.9

A subset $C \subseteq \mathbb{P}$ of a poset is *centered* iff for all $n \in \omega$ and $p_1, \ldots, p_n \in C$, there is a $q \in \mathbb{P}$ such that $q \leq p_i$, for $1 \leq i \leq n$. $\mathbb{P}$ is σ-centered iff $\mathbb{P} = \bigcup_{i \in \omega} C_i$, where each C_i is centered.

We can now state the main reason why we care about such posets

Theorem 5.2

Every σ-centered poset has the ccc.

Proof. In fact, since every $p, q \in C_i$ are compatible, each antichain contains at most one element of each C_i, but they form a countable partition of $\mathbb{P}$, so our antichain should also be countable. ■

Let's now finish our previous proof.

Lemma 5.6

Hechler Forcing has the ccc.

Proof. The more naive approach would be: If $\mathcal{A}$ is any uncountable antichain, since $\omega^{<\omega}$ is countable, by Pigeonhole Principle, there would have $(s, F), (s, H) \in \mathcal{A}$, but then $(s, F \cup H) \leq (s, F), (s, H)$, i.e., $(s, F) \not\perp (s, H)$, contradicting $\mathcal{A}$ is an antichain.

In the more general sceneario, we are actually proving $\mathbb{P}$ is σ-centered, i.e.,

$$\mathbb{P} = \bigcup_{p \in \omega^{<\omega}} C_p, \text{ where } C_p = \{(p, F) : F \in [\mathcal{F}]^{<\omega}\}$$

where each C_p is clearly centered. ■

5.4.3 MAD Families

Definition 5.10

$A, B \in [\omega]^\omega$ are called *almost disjoint* if $A \cap B$ is finite. A family $\mathcal{A} \subseteq [\omega]^\omega$ is called a *Maximal Almost Disjoint* (MAD) family if its elements are pairwise almost disjoint, and it's maximal. The almost disjoint number $\mathfrak{a}$ is the least cardinality of a MAD family.

Before analyzing such cardinal, let's guarantee it's well defined, i.e., that there exists an uncountable MAD family.

Lemma 5.7

Let $A \subseteq \omega$ be any infinite set, then there is an almost disjoint family $\{A_\alpha : \alpha < \mathfrak{c}\}$ of subsets of A, where almost disjoint means that for any $\alpha \neq \beta < \mathfrak{c}$, we have that $|A_\alpha \cap A_\beta| < \aleph_0$.

Proof. Since $A \subseteq \omega$ is infinite, there is a bijection $f : \mathbb{Q} \to A$, then it's sufficient to find an almost disjoint family $\{A_\alpha : \alpha < \mathfrak{c}\}$ of subsets of $\mathbb{Q}$. Then, for each irrational number α, let $\{q_n^\alpha\}_{n\in\omega} \subseteq \mathbb{Q}$ be a sequence converging to α, and let $A_\alpha = \{q_n^\alpha : n \in \omega\}$. Then, since α is irrational, each A_α is infinite and, if $\alpha \neq \beta$, $q_n^\alpha = q_n^\beta$ only for finitely many n's, so $\{f(A_\alpha) : \alpha < \mathfrak{c}\}$ is our family. ∎

Now to construct a MAD family we simply apply Zorn's Lemma in the expected way

Theorem 5.3

Every almost disjoint family $\mathcal{A}$ can be extended to a MAD family $\widetilde{\mathcal{A}}$.

Proof. Consider the following set ordered by inclusion:

$$\mathcal{F} = \{\mathcal{B} : \mathcal{B} \text{ is a MAD family and } \mathcal{A} \subseteq \mathcal{B}\}$$

Since $\mathcal{A} \in \mathcal{F}$, then it's non-empty. Now, given a chain C in $\mathcal{F}$, let $U = \bigcup C$, clearly, if $A, B \in U$, then $A, B \in \mathcal{B}$ for some $\mathcal{B} \in C$, and then A and B are almost disjoint, so U is a upper bound for C. Then, by Zorn's Lemma, $\mathcal{F}$ admits a maximal element $\widetilde{\mathcal{A}}$. ∎

We will now do a similar construction to show the following.

Theorem 5.4

Let $\kappa \geq \aleph_0$. Under MA(κ), we have $\mathfrak{a} > \kappa$.

Proof. Let $\mathcal{M}$ be an infinite MAD family, with $|\mathcal{M}| \leq \kappa$. We will create an $G \in [\omega]^\omega$ such that $G \cap A$ is finite, for every $A \in \mathcal{M}$, contradicting its maximality.

As in Hechler's Forcing, let's try the following forcing notion:

$$\mathbb{P} = \{(X, \mathcal{A}) : X \in [\omega]^{<\omega} \text{ and } \mathcal{A} \in [\mathcal{M}]^{<\omega}\}$$

And let's define $(X, \mathcal{A}) \leq (Y, \mathcal{B})$ iff $X \supseteq Y$, $\mathcal{A} \supseteq \mathcal{B}$, and $(X \setminus Y) \cap A = \emptyset$, for every $A \in \mathcal{A}$. It's routine to verify that $(\mathbb{P}, \leq)$ is a poset. Let's now define, for $n \in \omega$ and $A \in \mathcal{M}$

$$D_n = \{(X, \mathcal{A}) \in \mathbb{P} : \exists k \geq n\,[k \in X]\} \quad \text{and} \quad E_A = \{(X, \mathcal{A}) \in \mathbb{P} : A \in \mathcal{A}\}$$

Given $(X, \mathcal{A}) \in \mathbb{P}$, since $\mathcal{A}$ is finite and $\mathcal{M}$ infinite, let $C \in \mathcal{M} \setminus \mathcal{A}$. If $\omega \setminus \bigcup \mathcal{A}$ was finite, we would have $C \cap (\bigcup \mathcal{A})$ infinite and, by the Pigeonhole Principle, $C \cap A$ is infinite, for some $A \in \mathcal{A}$, contradicting that $\mathcal{M}$ is a MAD family. Then, $\omega \setminus \bigcup \mathcal{A}$ is infinite, and we can find some $k \geq n$, such that $k \notin \bigcup \mathcal{A}$. So $D_n \ni (X \cup \{k\}, \mathcal{A}) \leq (X, \mathcal{A})$. Similarly, $E_A \ni (X, \mathcal{A} \cup \{A\}) \leq (X, \mathcal{A})$.

We can also see that $\mathbb{P}$ is σ-centered, since we can write

$$\mathbb{P} = \bigcup_{X \in [\omega]^{<\omega}} C_X, \text{ where } C_X = \{(X, \mathcal{A}) : \mathcal{A} \in [\mathcal{M}]^{<\omega}\}$$

where each C_X is clearly centered. Then $\mathbb{P}$ has the ccc. Now, let

$$\mathcal{D} = \{D_n, E_A : n \in \omega,\ A \in \mathcal{M}\}$$

Then $|\mathcal{D}| \leq \omega \cdot \kappa = \kappa$, so, by MA($\kappa$), there is a $\mathcal{D}$-generic filter G. Let's define

$$X_G = \bigcup \{X : \exists \mathcal{A}\, (X, A) \in G\}$$

Since for every n, $G \cap D_n \neq \emptyset$, it means we have $k \in X_G$ for some $k \geq n$, so $X_G \in [\omega]^\omega$. Now, given $A \in \mathcal{M}$, let $(X, \mathcal{A}) \in G \cap E_A$, we claim that $(X_G \setminus \max X) \cap A = \emptyset$. In fact, let $n > \max X$, and $(Y, \mathcal{B}) \in G \cap D_n$. Since G is a filter, there is some $(Z, \mathcal{C}) \leq (X, \mathcal{A}), (Y, \mathcal{B})$, so $n \notin A$. ■

We could avoid the stinky $\max X$ by changing $Y \subseteq X$ in the $\leq$ definition with $Y \subseteq_e X$ (i.e., initial segment)

5.4.4 Amoeba Forcing

Definition 5.11

The additivity of null sets, called $\mathbf{add}(\mathcal{N})$, is the least amount of null sets whose union is not a null set.

Since this is the least cardinal in Cichoń's Diagram, the following Theorem shows that MA collapses all of Cichoń's Diagram to $\mathfrak{c}$.

Theorem 5.5

Let $\kappa \geq \aleph_0$, then MA(κ) implies $\mathbf{add}(\mathcal{N}) > \kappa$.

Proof. Given our null sets $\{N_\alpha\}_{\alpha<\kappa}$, we will show $N = \bigcup_{\alpha<\kappa} N_\alpha$ is also a null set by showing that, for all $\varepsilon > 0$, there is some open set $U_G \supseteq N$ such that $\mu(U_G) \leq \varepsilon$.

Let's then approximate it by $\mathbb{P} = \{U \subseteq \mathbb{R} : U \text{ is open and } \mu(U) < \varepsilon\}$, under reverse inclusion. Clearly our dense sets would be $D_\alpha = \{U \in \mathbb{P} : N_\alpha \subseteq U\}$, then, if all conditions to apply MA(κ) are satisfied, we would have a $\mathcal{D}$-generic filter G, where $\mathcal{D} = \{D_\alpha\}_{\alpha<\kappa}$, and we can let $U_G = \bigcup G$.

To show $\mu(U_G) \leq \varepsilon$, since $U_G \subseteq \mathbb{R}$, it's second countable and, therefore, Lindelöf, so the cover G of U_G admits a countable subcover $\{G_n\}_{n\in\omega} \subseteq G$. Since G is a filter, for any $G_0, \dots, G_k$, we can find a $\mathbb{P} \ni V \leq G_0, \dots, G_n$, so $\mu\left(\bigcup_{i=0}^k G_i\right) < \varepsilon$ and, therefore, $\mu(U_G) \leq \varepsilon$, as desired.

It then remains to show that $(\mathbb{P}, \supseteq)$ is ccc and D_α are dense. To see density, given $U \in \mathbb{P}$, fix $\alpha < \kappa$; since $\varepsilon - \mu(U) > 0$ and $\mu(N_\alpha) = 0$, we can find some open $V \supseteq N_\alpha$ such that $\mu(V) < \varepsilon - \mu(U)$, but then we have $W = U \cup V \leq U$ such that $\mu(W) < \varepsilon$ and $W \in D_\alpha$.

Finally, to prove ccc, the ideia is to use the countable basis $\mathcal{B}$ in $\mathbb{R}$ of finite union of intervals with rational endpoints. In particular, for any $U \in \mathbb{P}$ we can find some $V_U \in \mathcal{B}$, with $V_U \subseteq U$, such that $\mu(U \setminus V_U)$ is arbitrarily small. Let then it be such that $\mu(U \setminus V_U) < \frac{1}{2}(\varepsilon - \mu(U)) \leq \frac{1}{2}(\varepsilon - \mu(V_U))$. Now, let $\mathcal{A}$ be an uncountable antichain. But, since $\mathcal{B}$ is countable, we will have two distinct $U, W \in \mathcal{A}$ such that $V = V_U = V_W$, but then

$$\mu(U \cup W) \leq \mu(U \setminus V) + \mu(W \setminus V) + \mu(V) < \varepsilon$$

and $U \cup W \leq U, W$, i.e., $U \not\perp W$, contradicting that $\mathcal{A}$ is an antichain. ■

Such forcing notion if called Amoeba Forcing cause of V in the proof that each D_α is dense, where W is the amoeba, U is it nucleus, and V is the pseudopod it uses to reach the food N_α.

5.4.5 Mathias Forcing

Pendent

CHAPTER 6

Forcing

6.1 Basic Idea

6.1.1 Failure of Inner Models

Let's first see why we can't use the method of inner models to prove, for example, that $\neg$CH.

If we would follow the analogy in subsection 3.1, we could argue for example that if Φ proves there is some inner model M^G such that $M^G \vDash \neg\varphi$, we would have a problem, since in E^G we will have that $M^{E^G} \leq E^G$ and, therefore, $M^{E^G} = \{e_G\}$, by minimality. Therefore $M^{E^G} \vDash \varphi$, a contradiction. Then φ would be an example of a formula that we can't prove using the method of inner models.

In fact, we have the technical limitation that Group Theory is clearly not strong enough to define what is a formula or construct a satisfiability relation, but that's the intuitive idea.

> **Observation:** If we could define in ZF a transitive proper class $M \vDash \mathrm{ZFC} + \neg\mathrm{CH}$, then $\neg\,\mathrm{Con}(\mathrm{ZF})$.

Proof. If we could define such M in ZF, we could also define it in ZFC$+V = L$. Since M is a proper class model, by absoluteness of rank (Corollary **??**), we have that $\mathrm{ON} \subseteq M$. We also have that $L \subseteq M$ since, by Lemma 3.6, $L(\delta)$ is absolute, and, therefore, $M = L$ by $V = L$, so $M \vDash V = L$ and, therefore, $M \vDash$ CH, a contradiction. ■

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In other words, since L is the minimal model, we can't use the method of inner models to prove the relative consistency of the negation of any corollary of $V = L$.

With this in mind, to produce a model N for ZFC $+ \neg$CH, we need to step outside of our universe and create some $N \supset V$ that has sufficiently many subsets of ω so that CH fails in it.

6.1.2 Sketch of Forcing

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6.1.3 Finitistic Consistecy Proofs

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6.2 Generic Extensions

Throughout this subsection $\mathbb{P} \in M$ abbreviates $(\mathbb{P}, \mathbb{1}, \leq) \in M$, and we say G is $\mathbb{P}$-generic over M when G is a filter on $\mathbb{P}$ and $\mathcal{D}$-generic, where $\mathcal{D} = \{D \in M : D \text{ is dense in } \mathbb{P}\}$. Whenever M is a ctm, we have $\mathcal{D}$ countable and, by Lemma 5.2, it's clear that:

Lemma 6.1

(Rasiowa–Sikorski) If M is a ctm for $\mathrm{ZF} - \mathcal{P}$ and $\mathbb{P} \in M$, then for every $p \in \mathbb{P}$ there is a $\mathbb{P}$-generic G over M, with $p \in G$.

In the proof of the Generic Filter Existence Lemma, the notion of being dense is absolute, so this is not a problem. However, the enumeration of the dense sets $\mathcal{D}$ in order type ω must take place outside of M, since M itself doesn't know it's countable, so G is usually not in M:

Lemma 6.2

If $\mathbb{P} \in M$ is atomless and G is $\mathbb{P}$-generic over M, then $G \notin M$.

Proof. As in the proof of Corollary 5.3, $D = \mathbb{P} \setminus G$ is dense. If $G \in M$, then $D \in M$, contradicting that $G \cap D \neq \emptyset$. ■

Since we want to extend M to $M[G]$, as the set of all sets that can be built from G and M using "simple set-theoretic processes", if $G \in M$ then $M[G]$ is simply M, so posets with atom are of little interest in forcing.

Definition 6.1

τ is a $\mathbb{P}$-name iff τ is a relation and

$$\forall \langle \sigma, p \rangle \in \tau \, [\sigma \text{ is a } \mathbb{P}\text{-name} \wedge p \in \mathbb{P}]$$

Then $V^{\mathbb{P}}$ is the class of all $\mathbb{P}$-names.

As a side note, it's worth noting that the way we formally define a $\mathbb{P}$-name is by recursion on the well-founded relation xRy iff $x \in \operatorname{trcl}(y)$, where we let $F : V \to \{0, 1\}$ be given by: $F(\tau) = 1$ iff τ is a relation and $\forall \langle \sigma, p \rangle \in \tau \, [F(\sigma) = 1 \wedge p \in \mathbb{P}]$; and $F(\tau) = 0$ otherwise. Since all these notions are absolute for transitive models of $\mathrm{ZF} - \mathcal{P}$, then the notion of $\mathbb{P}$-name are absolute.

We will call $M^{\mathbb{P}}$ the set of $\mathbb{P}$-names in M, which is exactly $\{\tau \in M : (\tau \text{ is a } \mathbb{P} \text{ name})^M\}$.

Definition 6.2

If τ is a $\mathbb{P}$-name and $G \subseteq \mathbb{P}$, then define by recursion

$$\operatorname{val}(\tau, G) = \tau_G := \{\sigma_G : \exists p \in G\, [\langle \sigma, p\rangle \in \tau]\}$$

And let $M[G] = \{\tau_G : \tau \in M^{\mathbb{P}}\}$, whenever M is a transitive model of $\mathrm{ZF} - \mathcal{P}$, with $\mathbb{P} \in M$.

So, if τ represents a name, then τ_G represents its actual value. In some sense, τ is then a "fuzzy" representation of some actual set τ_G that is determined by G (or, way better, filtered by it). In fact, since $M^{\mathbb{P}} \in M$, then M can talk about names, but since in general $G \notin M$, then it cannot talk about its actual values. It's also not that hard to see that val is absolute.

With this in mind, let's try to give name to some known sets

Definition 6.3

For a forcing poset $(\mathbb{P}, \leq, \mathbb{1})$ and any set x, let $\check{x} = \{\langle \check{y}, \mathbb{1}\rangle : y \in x\}$.

This is, in some sense, a natural definition, since $\mathbb{1}$ is the unique element we can guarantee to be in G. And, as expected, we have the following properties:

Lemma 6.3

If M is a transitive model of $\mathrm{ZF} - \mathcal{P}$, with $\mathbb{P} \in M$ and G a filter on $\mathbb{P}$, then

1. $\forall x \in M\, [\check{x} \in M^{\mathbb{P}} \wedge \check{x}_G = x]$;
2. $M \subseteq M[G]$.

Proof. 1. Clearly $\check{x}$ is a $\mathbb{P}$-name in M by absoluteness. Now, using $\in$-induction, we know $\check{\emptyset} = \{\langle \check{y}, \mathbb{1}\rangle : y \in \emptyset\} = \emptyset$, then $\check{\emptyset}_G = \emptyset$. Now, if $\check{y}_G = y$ for all $y \in x$, then clearly

$$\check{x}_G = \{\langle \check{y}, \mathbb{1}\rangle : y \in x\}_G = \{\check{y}_G : y \in x\} = \{y : y \in x\} = x$$

2. And this is immediate by 1., since $\check{x}_G = x \in M[G]$ for all $x \in M$. ■

As noted earlier, usually $G \notin M$, but we can cook up a name Γ for G in M as follows:

Corollary 6.1

Let $\Gamma = \{\langle \check{p}, p\rangle : p \in \mathbb{P}\}$. Under the hypothesis of the previous lemma, we have that Γ is a $\mathbb{P}$-name and $\Gamma_G = G$. Hence $G \in M[G]$.

Proof. Clearly Γ is a $\mathbb{P}$-name and, by the previous lemma, we have that

$$\Gamma_G = \{\check{p}_G : \exists p \in G\, [\langle \check{p}, p\rangle \in \Gamma]\} = \{\check{p}_G : p \in G\} = \{p : p \in G\} = G$$

■

Let's now see which axioms $M[G]$ does satisfies under the hypothesis of the previous lemma.

Lemma 6.4

Under the hypothesis of the previous lemma, $M[G] \vDash \mathcal{E}, \mathcal{R}, \mathcal{U}$ and it's transitive.

Proof. Given $\tau_G \in M[G]$, if $x \in \tau_G$, then $x = \sigma_G$ for some $\sigma \in \operatorname{dom}(\tau)$, which means σ is a $\mathbb{P}$-name and, therefore, $\sigma_G = x \in M[G]$, then it's transitive.

By transitivity, Lemma 2.1 guarantees that $M[G] \vDash \mathcal{E}$, and $\mathcal{R}$ is true since we are under it.

Now, instead of $\mathcal{U}$ as it's usually defined, we will change it by an equivalent form (only under $\mathcal{C}$ or $\mathcal{F}$), which is that for any $x \in M[G]$, there is some $y \in M[G]$ such that $\bigcup x \subseteq y$, since Comprehension (when proved) will do the hard work to then find $\bigcup x$ inside of y.

For this, let $\tau_G \in M[G]$, then take $\sigma = \bigcup \operatorname{dom}(\tau)$. Since $\operatorname{dom}(\tau)$ only has $\mathbb{P}$-names, by the recursive definition we know their union has elements of the form $\langle \pi, p \rangle$, where π is a $\mathbb{P}$-name and $p \in \mathbb{P}$, so σ is also a $\mathbb{P}$-name, we claim $y = \sigma_G$. To see this, if $a \in b \in \tau_G$, then $b = \pi_G$ for some $\pi \in \operatorname{dom}(\tau)$, so $\pi \subseteq \bigcup \operatorname{dom}(\tau) = \sigma$, and $b = \pi_G \subseteq \sigma_G = y$, therefore, since $a \in b$, we have $a \in y$. ■

Let's now see where the concept of a filter being generic becomes important. We aim to prove that $M[G] \vDash \mathrm{ZF}$, in particular, since $G, \mathbb{P} \in M[G]$, then we would like that $C = \mathbb{P} \setminus G \in M[G]$ too, but what would be its name $\mathring{C}$? Since we have no access to G, we can take $\mathring{C}$ based on some property of the filter G. In particular, since they are all pairwise compatible, let $\mathring{C} = \{\langle \check{p}, q \rangle : p, q \in \mathbb{P} \wedge p \perp q\}$. Then $\mathring{C}_G = \{\check{p}_G : p \in \mathbb{P} \wedge \exists q \in G\, [p \perp q]\} = \{p \in \mathbb{P} : \exists q \in G\, [p \perp q]\}$, so if $p \in G$, then $\nexists q \in G$ incompatible with it, then $\mathring{C}_G \cap G = \emptyset$, but why $\mathring{C}_G \cup G = \mathbb{P}$?

In fact, if we merely assumed that G is a filter, then $\mathring{C}$ need not work, for example, if $G = \{\mathbb{1}\}$, then $\mathring{C}_G = \emptyset$. Furthermore, this would not be a problem of the name $\mathring{C}$ we choose, we can show that for mere filter G, the complement $\mathbb{P} \setminus G$ need not be in $M[G]$ at all. But, if G was generic... how to make it appear naturally?

Although $M[G]$ is bigger than M, the following lemma shows it's not too much bigger.

Lemma 6.5

Under the hypothesis of Lemma 6.2:

1. $\operatorname{rank}(\tau_G) \leq \operatorname{rank}(\tau)$ for all $\tau \in M^{\mathbb{P}}$;
2. $o(M[G]) = o(M)$;
3. $|M[G]| = |M|$.

Proof. (1) Clearly it's true for $\tau = \emptyset$. Now, assume it's true for all $\sigma \in \operatorname{dom}(\tau)$. Since

$$\operatorname{rank}(\tau) = \sup\{\operatorname{rank}(\langle \sigma, p \rangle) + 1 : \langle \sigma, p \rangle \in \tau\} \geq \operatorname{rank}(\langle \sigma, p \rangle) > \operatorname{rank}(\sigma)$$

then, we have that

$$\begin{aligned}
\operatorname{rank}(\tau_G) &= \sup\{\operatorname{rank}(\sigma_G) + 1 : \exists p \in G\, [\langle \sigma, p \rangle \in \tau]\} \\
&\leq \sup\{\operatorname{rank}(\sigma_G) + 1 : \sigma \in \operatorname{dom}(\tau)\} \\
&\leq \sup\{\operatorname{rank}(\sigma) + 1 : \sigma \in \operatorname{dom}(\tau)\} \\
&\leq \operatorname{rank}(\tau);
\end{aligned}$$

(2) Since $M \subseteq M[G]$, clearly $M \cap ON \subseteq M[G] \cap ON$. Now, given $\alpha \in M[G] \cap ON$, then $\alpha = \tau_G$ for some $\tau \in M^{\mathbb{P}}$, and, by the previous item, $\alpha = \mathrm{rank}(\alpha) = \mathrm{rank}(\tau_G) \leq \mathrm{rank}(\tau)$, but since $\mathbb{P}$-names and rank are absolute, then $\mathrm{rank}(\tau) \in M \cap ON$, so $\alpha \in M \cap ON$ and, therefore, $M \cap ON = M[G] \cap ON$;

(3) Since $M^{\mathbb{P}} \subseteq M \subseteq M[G]$, then

$$|M[G]| \leq |M^{\mathbb{P}}| \leq |M| \leq |M[G]|$$

■

It's almost a miracle that $M[G]$ is, in fact, the minimal transitive model of $\mathrm{ZF} - \mathcal{P}$ (as we will see) that contains both the elements of M and G:

Lemma 6.6

Under the hypothesis of Lemma 6.2, if $N \vDash \mathrm{ZF} - \mathcal{P}$ is transitive, with $M \subseteq N$ and $G \in N$, then $M[G] \subseteq N$.

Proof. Since val is absolute, if $\tau, G \in N$, then $\tau_G \in N$. ■

6.3 The Forcing Relation

In order to verify that $M[G] \vDash \mathrm{ZFC}$, the hardest axiom is $\mathcal{C}$. To illustrate the problem, let $\varphi(x, y)$ be a formula and $\sigma \in M^{\mathbb{P}}$, why would $S = \{n \in \omega : (\varphi(n, \sigma_G))^{M[G]}\} \in M[G]$? We need to somehow cook up a name τ for S for φ arbitrary, and we will see that our name will be something like

$$\tau = \{\langle \check{n}, p\rangle : n \in \omega \wedge p \in \mathbb{P} \wedge p \Vdash \varphi(\check{n}, p)\} \tag{3}$$

where $\Vdash$ is the forcing relation, that roughly says $M[G] \vDash \varphi(\check{n}, \sigma)$ whenever G is generic and $p \in G$.

Definition 6.4

The $\mathbb{P}$ forcing language $\mathcal{FL}_{\mathbb{P}}$ is $\mathcal{L}^S$, where $S = \{\in\} \cup V^{\mathbb{P}}$.

Then we can formalize what does it means to $M[G] \vDash \varphi(\check{n}, \sigma)$, when σ is a $\mathbb{P}$-name:

Definition 6.5

Let $\psi \in \mathcal{FL}_{\mathbb{P}} \cap M$, then $M[G] \vDash \psi$ is interpreted as usual, with $\tau^{M[G]} = \tau_G$.

And, finally, we can now define the forcing relation:

Definition 6.6

Let M be a ctm for $\mathrm{ZF} - \mathcal{P}$, $\mathbb{P} \in M$ and $\psi \in \mathcal{FL}_{\mathbb{P}} \cap M$ be a sentence. Then $p \Vdash_{\mathbb{P},M} \psi$ iff $M[G] \vDash \psi$ for all $\mathbb{P}$-generic filters G over M, with $p \in G$.

The assumption that M is countable is to guarantee, by Rasiowa-Sirkoski, that such G exists. Note that it's a immediate corollary from the definition that: if $p \Vdash \varphi$ and $q \leq p$, then $q \Vdash \varphi$.

Now, using our τ defined in (3), we have two major problems: First, why is $\tau \in M^{\mathbb{P}}$? To define $\Vdash$ we considered not only one generic filter, but all of them. Second, why does $\tau_G = S$? The first problem is addressed by the Definability Lemma, which roughly says that forcing is definable in M, and the second by the Truth Lemma, which says that anything true in $M[G]$ is forced by something. Let's state them without proof by now:

Lemma 6.7

(Truth Lemma) Let M be a ctm for $\mathrm{ZF} - \mathcal{P}$, with $\mathbb{P} \in M$, $\psi \in \mathcal{FL}_{\mathbb{P}} \cap M$ a sentence and G a $\mathbb{P}$-generic filter over M. Then

$$M[G] \vDash \psi \iff \exists p \in G\,[p \Vdash \psi]$$

Now, naively, the Definability Lemma should be that $\{(p, \psi) : p \Vdash \psi\} \in \mathcal{D}^+(M)$, but, if $\mathbb{P} = \{\mathbb{1}\}$, then every name is of the form $\check{a}$, and $\mathbb{1} \Vdash \psi$ iff $M \vDash \psi$, which would contradict Tarski's Theorem on Undefinability of Truth. A correct statement is:

Lemma 6.8

(Definability Lemma) Let M be a ctm for $\mathrm{ZF} - \mathcal{P}$ and $\varphi(\overline{x}) \in \mathcal{L}^{\{\in\}}$, then

$$\left\{(p, \mathbb{P}, \leq, \mathbb{1}, \overline{\vartheta}) : p \in \mathbb{P} \wedge \mathbb{P} \in M \text{ is a forcing poset} \wedge \overline{\vartheta} \in M^{\mathbb{P}} \wedge p \Vdash \varphi(\overline{\vartheta})\right\} \in \mathcal{D}^-(M)$$

We can, in some sense, talk about the elements of $M[G]$ cooking $\mathbb{P}$-names in M. In a lot of these cases, the name we gonna cook up will use the forcing relation $\Vdash$, and here is where The Definability Lemma comes in, if we can express $\Vdash$ in M, we can do all our work inside M, which we know is a model of ZFC. Using such name τ in M, if $p \Vdash \varphi$ appears in it's definition, τ_G will have something like $\exists p \in G\,[p \Vdash \varphi]$, and then $M[G] \vDash \varphi$ by definition. Conversely, if we know $M[G] \vDash \varphi$, The Truth Lemma gives us the other direction.

With these lemmas in hand, let's prove that $M[G] \vDash \mathrm{ZFC}$, we will split this proof in two:

Lemma 6.9

Let M be a ctm for $\mathrm{ZF} - \mathcal{P}$, $\mathbb{P} \in M$ and G, $\mathbb{P}$-generic over M. Then $M[G] \vDash \mathcal{C}, \mathcal{I}$.

Proof. Let $\varphi(x, z, \overline{v}) \in \mathcal{L}^{\{\in\}}$ without y free, and $N = M[G]$; we must check that

$$\forall z, \overline{v} \in N\, \exists y \in N\, \forall x \in N\, [x \in y \leftrightarrow x \in z \wedge \varphi^N(x, z, \overline{v})]$$

Let $\pi_G, \overline{\sigma_G} \in N$ (corresponding to $z, \overline{v}$) and let

$$S = \{\vartheta \in \pi_G : \varphi^N(\vartheta, \pi_G, \overline{\sigma_G})\} = \{\vartheta_G : \vartheta \in \mathrm{dom}(\pi) \wedge N \vDash (\vartheta \in \pi \wedge \varphi(\vartheta, \pi, \overline{\sigma}))\}$$

then it's sufficient to prove that $S \in N$. Let's then cook a name for it, define:

$$\tau = \{\langle \vartheta, p \rangle : \vartheta \in \mathrm{dom}(\pi) \wedge p \in \mathbb{P} \wedge p \Vdash (\vartheta \in \pi \wedge \varphi(\vartheta, \pi, \overline{\sigma}))\}$$

then $\tau_G = \{\vartheta_G : \vartheta \in \mathrm{dom}(\pi) \wedge \exists p \in G\,[p \Vdash (\vartheta \in \pi \wedge \varphi(\vartheta, \pi, \overline{\sigma}))]\}$. By the Definability Lemma, $\tau \in M^{\mathbb{P}}$ and, if $\vartheta_G \in \tau_G$, then $p \Vdash (\vartheta \in \pi \wedge \varphi(\vartheta, \pi, \overline{\sigma}))$, in particular, since $p \in G$ and G is $\mathbb{P}$-generic over M, then $M[G] = N \vDash (\vartheta \in \pi \wedge \varphi(\vartheta, \pi, \overline{\sigma}))$, so $\vartheta_G \in S$ too.

Fix now $\vartheta_G \in S$, where $\vartheta \in \mathrm{dom}(\pi)$. Then $N \vDash (\vartheta \in \pi \wedge \varphi(\vartheta, \pi, \overline{\sigma}))$ and, by the Truth Lemma, there is a $p \in G$ such that $p \Vdash (\vartheta \in \pi \wedge \varphi(\vartheta, \pi, \overline{\sigma}))$, then $\langle \vartheta, p \rangle \in \tau$, so $\vartheta_G \in \tau_G$.

Now, to show $\mathcal{I}$, since $\omega \in M[G]$, it follows from Lemma 2.3. ■

Let's now prove that $M[G] \vDash$ ZFC in fact.

Theorem 6.1

If M be a ctm for ZF, $\mathbb{P} \in M$ and G a $\mathbb{P}$-generic filter over M, then $M[G] \vDash$ ZF. Furthermore, if $M \vDash$ ZFC, then $M[G] \vDash$ ZFC.

Proof. By the previous Lemmas we just need to show that $M[G] \vDash \mathcal{P}, \mathcal{F}, \mathrm{AC}$.

$\mathcal{P}$: Let $\tau_G \in M[G]$, we will find a set $b \in M[G]$ such that $\mathcal{P}(\tau_G) \cap M[G] \subseteq b$. So, working with τ in M, let's find a set big enough so it have names for the elements of $\mathcal{P}(\tau_G)$. Let $Q = (\mathcal{P}(\mathrm{dom}(\tau) \times \mathbb{P}))^M$, the set of all names $\vartheta \in M^{\mathbb{P}}$ such that $\mathrm{dom}(\vartheta) \subseteq \mathrm{dom}(\tau)$. Let then $\pi = Q \times \{\mathbb{1}\}$ so we have that $b = \pi_G = \{\vartheta_G : \vartheta \in Q\}$. Now, given $\theta_G \in \mathcal{P}(\tau_G) \cap M[G]$, let's find a name ϑ for θ_G in Q. Let

$$\vartheta = \{\langle \sigma, p \rangle : \sigma \in \mathrm{dom}(\tau) \wedge p \Vdash \sigma \in \theta\} \in Q$$

Then $\vartheta \in M$ by the Definability Lemma. So, let $\sigma_G \in \vartheta_G = \{\sigma_G : \exists p \in G\, [p \Vdash \sigma \in \theta]\}$, then $M[G] \vDash \sigma_G \in \theta_G$ and, therefore, $\vartheta_G \subseteq \theta_G$. Now, since $\theta_G \subseteq \tau_G$, then every element $\sigma_G \in \theta_G$ has $\sigma \in \mathrm{dom}(\tau)$ and, by the Truth Lemma, there is some $p \in G$ such that $p \Vdash \sigma \in \theta$, then $\langle \sigma, p \rangle \in \vartheta$ and, therefore, $\sigma_G \in \vartheta_G$.

$\mathcal{F}$: If $a \in M[G]$ and $(\forall x \in a\, \exists y\, \varphi(x, y))^{M[G]}$, then, since we now have comprehension, it's sufficient to show that there is a $b \in M[G]$ such that $(\forall x \in a\, \exists y \in b\, \varphi(x, y))^{M[G]}$. So, if $a = \tau_G$, working with τ in M let's find a set Q of names such that for all $p \in \mathbb{P}$ and all $\sigma \in \mathrm{dom}(\tau)$, if there is some name ϑ such that $p \Vdash \varphi(\sigma, \vartheta)$, then at least one such ϑ is in Q.

In order to obtain such Q, let's use the Reflection Theorem in M, with $A_\alpha = (R(\alpha))^M$ and (using the Definability Lemma) how the hell do I use Reflection's Theorem here?

Let then $\pi = Q \times \{\mathbb{1}\}$ and $b = \pi_G$. Now, if $\sigma_G \in a$, for $\sigma \in \mathrm{dom}(\tau)$, then $M[G] \vDash \exists y\, \varphi(\sigma, y)$ by hypothesis, so $M[G] \vDash \varphi(\sigma, \vartheta)$, for some $\vartheta \in M^{\mathbb{P}}$. Then, by the Truth Lemma, $p \Vdash \varphi(\sigma, \vartheta)$ for some $p \in G$ and, by definition of Q, there is some $\vartheta \in Q$ such that $p \Vdash \varphi(\sigma, \vartheta)$, then $y = \vartheta_G$ is such that $y \in b$ and $(\varphi(\sigma, \vartheta))^{M[G]}$.

AC: Let $\tau_G \in M[G]$. Working in M, well order $\mathrm{dom}(\tau)$ as $\{\sigma^\xi : \xi < \alpha\}$, define $\mathring{f} = \{\langle \mathrm{op}(\check{\xi}, \sigma^\xi), \mathbb{1} \rangle : \xi < \alpha\}$ and let $f = \mathring{f}_G = \{\langle \xi, \sigma^\xi_G \rangle : \xi < \alpha\} : \alpha \to A$, where $\tau_G \subseteq A$. Then, we can well-order τ_G in $M[G]$ as: $x \lhd y$ iff $\min\{\xi < \alpha : f(\xi) = x\} < \min\{\xi < \alpha : f(\xi) = y\}$. Pendent ■

It's worth noting that, by now, we explained why do we need objects like generic filters G to construct our new model $M[G]$ with the desired properties, but we haven't explained the importance of the ccc condition. This will be clear in subsection 6.5, where we relate the ccc condition with cardinal arithmetic in our new model $M[G]$.

6.4 Proof of Truth and Definability Lemmas

The aim of this subsection is to prove The Truth Lemma 6.3 and The Definability Lemma 6.3.

6.4.1 Motivational Properties

In order to define $\Vdash$ in M, we will do the reverse work, let's use both lemmas to see some properties it satisfies and then define another forcing relation $\Vdash^*$ in M based on those properties. It then remains to see they are sufficient so that $\Vdash$ and $\Vdash^*$ coincide in M.

Lemma 6.10

For any forcing poset $\mathbb{P} \in M$ and sentences $\varphi, \psi \in \mathcal{FL}_{\mathbb{P}} \cap M$:

1. No p force both φ and $\neg\varphi$;
2. If φ, ψ are logically equivalent, then $p \Vdash \varphi$ iff $p \Vdash \psi$;
3. If $p \Vdash \varphi$ and $q \leq p$, then $q \Vdash \varphi$;
4. $p \Vdash \varphi \wedge \psi$ iff $p \Vdash \varphi$ and $p \Vdash \psi$;
5. $p \Vdash \neg\varphi$ iff $\nexists q \leq p\, [q \Vdash \varphi]$;
6. Other connectives that follows from 4 and 5

Proof. 1. Since M is a ctm, there is always a generic G, with $p \in G$. Then clearly we cannot have $M[G] \vDash \varphi$ and $M[G] \vDash \neg\varphi$;

2. $p \Vdash \varphi$ iff $M[G] \vDash \varphi$ for all generic G, with $p \in G$, iff $M[G] \vDash \psi$ (since they are logically equivalent), iff $p \Vdash \psi$, as desired;

3. If G is generic with $q \in G$, then, since it's a filter, and $q \leq p$, then $p \in G$ and, therefore, $p \Vdash \varphi$ implies that $M[G] \vDash \varphi$, so $q \Vdash \varphi$;

4. Trivially follows from $N \vDash \varphi \wedge \psi$ iff $N \vDash \varphi$ and $N \vDash \psi$;

5. ($\Rightarrow$) If $q \Vdash \varphi$, for some $q \leq p$, by items (3) $p \Vdash \neg\varphi$ would imply that $q \Vdash \neg\varphi$, contradicting (1);

($\Leftarrow$) If $p \nVdash \neg\varphi$, then there is some generic G, with $p \in G$, such that $M[G] \vDash \varphi$. By the Truth Lemma, there is some $r \in G$ such that $r \Vdash \varphi$ and, since it's a filter, let $q \leq r, p$, then by item (3) we have that $q \Vdash \varphi$.

6. Pendent ■

In fact, the same argument we used to prove item (5) yields:

Lemma 6.11

If G is $\mathbb{P}$-generic over M, and $M[G] \vDash \varphi$, with $p \in G$, then there is a $q \in G$ such that $q \leq p$ and $q \Vdash \varphi$.

Let's now see how $\Vdash$ behaves with quantification:

Lemma 6.12

For any forcing poset $\mathbb{P} \in M$ and $\varphi(x) \in \mathcal{FL}_{\mathbb{P}} \cap M$:

1. $p \Vdash \forall x\, \varphi(x)$ iff $p \Vdash \varphi(\tau)$ for all $\tau \in M^{\mathbb{P}}$;
2. Same, but for $\exists$, which follows by the previous lemmas

Proof. 1. Is also trivial by definition, $p \Vdash \forall x\, \varphi(x)$ iff $M[G] \vDash \forall x\, \varphi(x)$ for all generic G, with $p \in G$, which is equivalent to $M[G] \vDash \varphi(\tau)$ for all $\tau \in M^{\mathbb{P}}$ and G generic containing p.
2. Pendent ∎

Pendent

6.4.2 Definition of $\Vdash^*$

Pendent

6.5 Cardinal Collapse

Throughout this subsection, let M be a ctm for ZFC. In order to discuss cardinal arithmetic in $M[G]$, let's pin down what the cardinal are. Although M and $M[G]$ have the same ordinals, they need not have the same cardinals, as we will see.

In fact, let $\mathbb{P} = \mathrm{Fn}(I, J)$, with $I, J \in M$ infinite, and let G be $\mathbb{P}$-generic over M, with $f = \bigcup G$. Then $f \in M[G]$ and, as we saw in Lemma 5.2, since all those dense sets D_i, R_j and E_h are in M by absoluteness, f is a function from I onto J.

Consider then $I = \omega$ and $J = \omega_5^M$, then $\omega^M = \omega$ by absoluteness, but ω_5^M is some countable ordinal, although M believes that it's the fifth uncountable cardinal. But $M[G]$ contains a map f from ω onto ω_5^M, so $M[G]$ knows that ω_5^M is countable, this phenomena is called cardinal collapse.

Definition 6.7

For a forcing poset $\mathbb{P} \in M$:

1. $\mathbb{P}$ preserves cardinals iff for all generic G:
$$(\beta \text{ is a cardinal})^M \text{ iff } (\beta \text{ is a cardinal})^{M[G]}, \text{ for all } \beta < o(M)$$
2. $\mathbb{P}$ preserves cofinality iff for all generic G:
$$\mathrm{cf}^M(\gamma) = \mathrm{cf}^{M[G]}(\gamma), \text{ for all limit } \gamma < o(M)$$

Note that $\mathrm{cf}^M(\gamma) \geq \mathrm{cf}^{M[G]}(\gamma)$, for all limit $\gamma < o(M)$, since we have that

$$\mathrm{cf}(\gamma) = \min\{\mathrm{ot}(X) : X \subseteq \gamma \wedge \sup(X) = \gamma\}$$

and, by absoluteness of $\mathrm{ot}(X)$, if $X \in M$ witnesses $\mathrm{cf}^M(\gamma)$, then $X \in M[G]$ also guarantees us that $\mathrm{cf}^{M[G]}(\gamma) \leq \mathrm{ot}^{M[G]}(X) = \mathrm{ot}^M(X) = \mathrm{cf}^M(\gamma)$.

The next lemma says that, to guarantee $\mathbb{P}$ preserves cardinals, it's sufficient to study when all limit β remains regular from M to $M[G]$, with $\omega < \beta < o(M)$.

Lemma 6.13

For a forcing poset $\mathbb{P} \in M$:

1. $\mathbb{P}$ preserves cofinalities iff for all generic G:

$$(\beta \text{ is regular})^M \to (\beta \text{ is regular})^{M[G]}, \text{ for all limit } \beta \text{ with } \omega < \beta < o(M)$$

2. If $\mathbb{P}$ preserves cofinalities, then $\mathbb{P}$ preserves cardinals.

Proof. 1. $(\Rightarrow)$ is trivial.

$(\Leftarrow)$ Pendent

2. Given a cardinal κ, either $\kappa \leq \omega$, and then we know both M and $M[G]$ have it, or κ is regular, and we saw they're the same in M and $M[G]$, or κ is singular, then $\kappa = \aleph_\eta$ for some limit η. So

$$\sup\{\aleph_\xi : \xi < \eta\} = \aleph_\eta = \sup\{\aleph_{\xi+1} : \xi < \eta\}$$

and, since every successor cardinal is regular, the singular cardinals are the same. ∎

It remains to find a good condition for when $\mathbb{P}$ does preserves cofinalities, and here is where the condition that $\mathbb{P}$ is ccc appears as an important concept.

Theorem 6.2

If $\mathbb{P} \in M$ and $(\mathbb{P} \text{ is ccc})^M$, then $\mathbb{P}$ preserves cofinalities and, hence, cardinals too.

In order to prove such theorem, we need the following "approximation lemma", where if $f : A \to B$ in $M[G]$, then f is not necessarily in M, but we can, within M, pick out a countable set of possible values for each $f(a)$.

Lemma 6.14

Let $\mathbb{P} \in M$, $(\mathbb{P} \text{ is ccc})^M$, and $A, B \in M$. If G is $\mathbb{P}$-generic over M and $f \in M[G]$, with $f : A \to B$, then there is an $F : A \to \mathcal{P}(B)$ with $F \in M$ such that for all $a \in A$, $f(a) \in F(a)$ and $(|F(a)| \leq \aleph_0)^M$.

Proof. Let $\mathring{f}_G = f$, then $\varphi \equiv (\mathring{f} : \check{A} \to \check{B}) \in \mathcal{FL}_{\mathbb{P}} \cap M$ is true in $M[G]$ and, by the Truth Lemma, there is some $p \in G$ such that $p \Vdash \varphi$. Now, let $F : A \to \mathcal{P}(B)$ be such that

$$F(a) = \{b \in B : \exists q \leq p \, [q \Vdash \mathring{f}(\check{a}) = \check{b}]\}$$

Then $F \in M$ by the Definability Lemma.

To see $f(a) \in F(a)$: Let $b = f(a)$, then $M[G] \vDash \mathring{f}(\check{a}) = \check{b}$, so by Lemma 6.4.1, there is a $q \leq p$ in G such that $q \Vdash \mathring{f}(\check{a}) = \check{b}$, then $b \in F(a)$.

To see $(|F(a)| \leq \aleph_0)^M$: Given $b \in F(a)$, let $q_b \leq p$ be such that $q_b \Vdash \mathring{f}(\check{a}) = \check{b}$ then, since $\mathring{f}_G$ is a function, q_b are pairwise incompatible. Furthermore, we can let $b \mapsto q_b$ be in M using the Definability Lemma and AC^M. Now, in M, since $b \neq c \Rightarrow q_b \perp q_c$, then by ccc $|F(a)| \leq \aleph_0$. ∎

Let's now proceed to the proof of Theorem 6.5

Proof. Let $\omega < \beta < o(M)$ a limit ordinal, with $(\beta \text{ is regular})^M$. If β is not regular in $M[G]$, then there is some $X \in M[G]$ cofinal in β such that $\alpha := \mathrm{ot}(X) < \beta$. Let $f : \alpha \to X$ be the order preserving bijection, then $f : \alpha \to \beta$ and there is an $F : \alpha \to \mathcal{P}(\beta)$ as in the previous lemma. Let $Y = \bigcup_{\xi<\alpha} F(\xi)$, then $Y \subseteq \beta$ and $\sup(Y) = \beta$.
So, within M, $Y = \sup_{\xi<\alpha} F(\xi)$ is a union of $\alpha < \beta$ countable sets, then $|Y| < \beta$, so we have that $\mathrm{cf}(\beta) \leq |Y| < \beta$, contradicting that β is an uncountable regular cardinal. ■

6.6 Models for ZFC + ¬CH

We now have all the tools to construct a model of ZFC + ¬CH.

Lemma 6.15

If $\alpha < o(M)$, $\kappa = (\aleph_\alpha)^M$, $\mathbb{P} = \mathrm{Fn}(\kappa \times \omega, 2)$ and G be $\mathbb{P}$-generic over M, then

$$M[G] \vDash 2^{\aleph_0} \geq \aleph_\alpha$$

Proof. $(\mathbb{P} \text{ is ccc})^M$ by Lemma 5.2 so, by Theorem 6.5 we also have $\kappa = (\aleph_\alpha)^{M[G]}$. Furthermore, as in the proof of Lemma 5.2, $f_G = \bigcup G : \kappa \times \omega \to 2$, which we can think of as a family of functions $h_\alpha : \omega \to 2$, for each $\alpha < \kappa$, where $h_\alpha(n) = f_G(\alpha, n)$. Now, for $\alpha < \beta < \kappa$, the dense sets

$$E_{\alpha,\beta} = \{p \in \mathbb{P} : \exists n \, [(\alpha, n), (\beta, n) \in \mathrm{dom}(p) \wedge p(\alpha, n) \neq p(\beta, n)]\}$$

are also in M, which means G meets these too, so h_α are all different functions.
Then $\{h_\alpha : \alpha < \kappa\}$ is a set of κ different elements of $\mathcal{P}(\omega)$, so $|\mathcal{P}(\omega)| = 2^{\aleph_0} \geq \kappa = (\aleph_\alpha)^{M[G]}$. ■

Let's now prove an really interesting theorem, sometimes referenced by Solovay's article '$2^{\aleph_0}$ can be anything it ought to be'. It say's $\mathfrak{c}$ can be exatly $\aleph_\alpha$, whenever this does not contradict König's Theorem, i.e., whenever $\aleph_\alpha > \omega$, with $\mathrm{cf}(\aleph_\alpha) > \omega$.

Definition 6.8

A nice name for a subset of $\tau \in V^{\mathbb{P}}$ is a name of the form

$$\bigcup\{\{\sigma\} \times A_\sigma : \sigma \in \mathrm{dom}(\tau)\}$$

where each A_σ is an antichain in $\mathbb{P}$.

Lemma 6.16

Let $\tau \in V^{\mathbb{P}}$ and $\mathbb{P}$ be ccc. Then, if $\kappa = |\mathbb{P}|, \lambda = |\mathrm{dom}(\tau)| \geq \omega$, there are no more than κ^λ nice names for subsets of τ.

Proof. Since $\mathbb{P}$ is ccc, there are no more than $|[\kappa]^{\leq\omega}| = \kappa^{\aleph_0}$ antichains in it, hence no more than $(\kappa^{\aleph_0})^\lambda = \kappa^{\aleph_0 \cdot \lambda} = \kappa^\lambda$ nice names for subsets of τ. ■

Let's now see every subset of τ is named by a nice name

Lemma 6.17

If $\mathbb{P} \in M$ and $\tau, \mu \in M^{\mathbb{P}}$, then there is a nice name $\vartheta \in M^{\mathbb{P}}$ for a subset of τ such that $\mathbb{1} \Vdash (\mu \subseteq \tau \to \mu = \vartheta)$.

Proof. Within M, given $\sigma \in \text{dom}(\tau)$, we can consider (using the Definability Lemma)

$$\mathcal{A}_\sigma = \{A \in \mathbb{P} : A \text{ is an antichain} \wedge \forall p \in A\, [p \Vdash \sigma \in \mu]\}$$

Then, $\mathcal{A}_\sigma \neq \emptyset$, since $\emptyset \in \mathcal{A}_\sigma$. Now, if C is a chain in $\mathcal{A}_\sigma$, then $A = \bigcup C$ is such that, if $p, q \in A$, then $p \in A_1$ and $q \in A_2$, for $A_1, A_2 \in C$. But, since C is a chain, assume WLOG that $A_1 \subseteq A_2$, then $p, q \in A_2$, which is an antichain, so $p \perp q$ and, therefore, A is also antichain.

Now, if $p \in A$, then $p \in A'$ for some $A' \in C$ and, by definition, $p \Vdash \sigma \in \mu$, so $A \in \mathcal{A}_\sigma$ is an upper bound for C. Now, by Zorn's Lemma, we know there is a maximum A_σ in $(\mathcal{A}_\sigma, \subseteq)$.

Let $\vartheta = \bigcup\{\{\sigma\} \times A_\sigma : \sigma \in \text{dom}(\tau)\}$. For a $\mathbb{P}$-generic filter G over M, assume $\mu_G \subseteq \tau_G$, then:

($\subseteq$): If $\sigma_G \in \vartheta_G$, with $\langle \sigma, p \rangle \in \vartheta$, for some $p \in G$ then, by definition of ϑ, $p \Vdash \sigma \in \mu$, so $\sigma_G \in \mu_G$;

($\supseteq$): If $\sigma_G \in \mu_G \setminus \vartheta_G$, since $\mu_G \subseteq \tau_G$, then we can let $\sigma \in \text{dom}(\tau)$. So $M[G] \vDash (\sigma \in \mu \wedge \sigma \notin \vartheta)$ and, by the Truth Lemma, let $q \in G$ be such that $q \Vdash \sigma \in \mu$ and $q \Vdash \sigma \notin \vartheta$, then, if $p \in A_\sigma$, we have $\langle \sigma, p \rangle \in \vartheta$, so $p \Vdash \sigma \in \vartheta$, contrary to $q \Vdash \sigma \notin \vartheta$. Then, by Lemma 6.4.1 (1) and (3), we have $p \perp q$, but $\langle \sigma, q \rangle \notin \vartheta$, then $q \notin A_\sigma$, contradicting its maximality. ■

We now have an upper bound:

Lemma 6.18

Assume M proves that: $\mathbb{P} \in M$ is ccc and $\kappa = |\mathbb{P}|, \lambda, \delta$ are infinite cardinals with $\delta = (\kappa^\lambda)^M$. If G be $\mathbb{P}$-generic over M, then $(2^\lambda \leq \delta)^{M[G]}$.

Proof. Within M: By Lemma 6.6, since $\check{\lambda} = \{\langle \check{\xi}, \mathbb{1} \rangle : \xi < \lambda\}$ has size λ, we can list all the nice names for subsets of $\check{\lambda}$ as $\{\vartheta_\zeta : \zeta < \delta\}$. Then, let $\mathring{f} = \{\langle \text{op}(\check{\zeta}, \vartheta_\zeta), \mathbb{1} \rangle : \zeta < \delta\}$.

Within $M[G]$: $\mathring{f}_G$ is a function with domain δ and $\mathring{f}_G(\zeta) = (\vartheta_\zeta)_G$. Furthermore, if $s \subseteq \lambda$, with $s \in M[G]$, then $s = \mu_G$ for some μ and, by Lemma 6.6, there is some ζ such that $\mathbb{1} \Vdash (\mu \subseteq \check{\lambda} \to \mu = \vartheta_\zeta)$, so $\mathring{f}_G(\zeta) = \mu_G$. Thus we have $\mathcal{P}(\lambda) \subseteq \text{Im}(\mathring{f}_G)$ and, therefore, $(2^\lambda \leq \delta)^{M[G]}$. ■

In particular, we can now prove the following:

Theorem 6.3

If $(\kappa = \aleph_\alpha \wedge \kappa^{\aleph_0} = \kappa)^M$, let $\mathbb{P} = \text{Fn}(\kappa \times \omega, 2)$ and let G be $\mathbb{P}$-generic over M, then

$$M[G] \vDash (2^{\aleph_0} = \aleph_\alpha = \kappa)$$

Proof. Since $\kappa = (\aleph_\alpha)^M$, Lemma 6.6 guarantees that $M[G] \vDash 2^{\aleph_0} \geq \aleph_\alpha$. Now, applying Lemma 6.6 for $\lambda = \aleph_0$ and $\kappa = |\mathbb{P}| = \aleph_\alpha$, we have $M[G] \vDash 2^{\aleph_0} \leq \delta = \kappa^{\aleph_0} = \kappa = \aleph_\alpha$. ■

As a corollary we can prove a really interesting result that says the continuum can consistently be any cardinal of uncountable cofinality.

Corollary 6.2

Assuming Con(ZFC), let κ be any cardinal with uncountable cofinality, then there is a ctm for ZFC where $\mathfrak{c} = \kappa$ and $\forall \lambda \geq \mathfrak{c}\, [2^\lambda = \lambda^+]$.

Proof. Since Con(ZFC), by Mostowski Collapsing Lemma there is a ctm N for ZFC, then let $M = L(o(N)) \vDash \mathrm{ZFC} + V = L$. Since $V = L \to \mathrm{GCH}$, which implies that $\mathrm{cf}(\kappa) > \omega$ iff $\kappa^{\aleph_0} = \kappa$, then, by the previous theorem, $M[G] \vDash (\mathfrak{c} = \kappa)$. Pendent ■

Add References